

RESEARCH ARTICLE

Multitask Deep Learning for Predicting Parkinson's Progression and Depression From Multimodal Time Series Data

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This research was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean Government [Ministry of Science and Information and Communication Technology (MSIT)] under Grant RS-2025-25398654.

ABSTRACT The high cases of Parkinson's disease (PD) in the senior generation require reliable PD progression prediction. PD can be associated with depression as a common non-motor symptom. Depression in Parkinson's disease (PD) is frequently underdiagnosed, despite its significant impact on patients' quality of life and caregiver burden. Early detection and identification of predictive factors are essential for timely intervention. This study proposes a novel multitask hybrid deep learning framework that leverages multimodal time-series data—including demographic, motor, and non-motor features—to enhance the detection and understanding of depressive symptoms in PD patients. These three modalities are early fused, and the relevant features of the resulting feature set were extracted using a deep variational autoencoder model. Because MRI data are critical in neurodegenerative disease diagnosis, a 3D CNN module has been used to learn deep features from MRI images, and these deep representations were lately fused with the extracted features of the variational autoencoder. The extracted deep representations were used to build a multitask classifier based on a novel Bidirectional LSTM model. This model has been optimized to predict the three tasks of 1) a four-class NHY scale that defines the severity of Parkinson's disease, 2) a binary class that determines whether or not a patient is depressed, and 3) a four-class depression scale that indicates the levels of depression. The proposed model has been evaluated on single-task and multitask problems with various time steps. Using a 2-time step in the NHY prediction task, our model obtains the maximum cross-validation and testing accuracies of 96.78% and 94.37%, respectively. The multitask model produced steady increases in testing accuracy of 86.29%, 87.12%, and 95.81% with 2-time step, 4-time step, and 6-time step, respectively. The proposed model investigated 1059 PD patients over six-time steps and 850 MRI images of single-time steps from the Parkinson's Progression Markers Initiative study (PPMI) dataset.

INDEX TERMS Parkinson's disease progression, depression, deep learning, Bi-LSTM, variational auto-encoders.

I. INTRODUCTION

The Parkinson's disease (PD) is a complex and progressive neurological disorder that has a major impact on quality of life, morbidity, and mortality [1]. Tremor, bradykinesia, stiffness, and other symptoms. Postural instability was long thought to be a motor indication of Parkinson's disease.

The associate editor coordinating the review of this manuscript and approving it for publication was Jinhua Sheng¹.

The etiology of PD extends beyond motor indications. As a result of this, Non-Motor Symptoms (NMS) coexist with [2] motor symptoms. There is evidence that the seriousness of NMS has a negative impact on people's quality of life. NMS has just been added to the list of PD symptoms such as olfactory dysfunction, pain, disturbances in sleep behavior, skin irregularities, and mental illness [3]. The most common neuropsychiatric problems, according to a study [4] are depression, anxiety, delusions, apathy, and aggressive

conduct. These NMS do not commonly report, and physicians do not address them [5]. Chaudhuri et al. [6] show 62% of NMS in patients go unnoticed by healthcare providers because patients are either humiliated or uninformed that the symptoms are related to the disease. The most common psychological symptom of PD is depression [7]. More than 50% of people having PD exhibit symptoms of depression [8], [9], and 10 to 20% have a major depressive disorder [9]. Changes in brain structure, neurotransmitters, and inflammatory and neurotrophic factors all have a role in depression in PD [10]. However, because of the NMS and motor symptoms associated with PD, depression is frequently misunderstood. The underlying processes of depression in PD are not fully understood, although alterations in brain structure, neurotransmitter signaling, and levels of inflammatory and neurotrophic substances are proposed to contribute to PD development [11]. Depression has a wide range of consequences, including earlier dopaminergic drug initiation, increased functional impairment, faster physical and cognitive decline, increased mortality, worse quality of life, and increased caregiver suffering [12]. In the end, it leads to wasteful social spending at the national level. Furthermore, individuals with depression had significantly greater anxiety symptoms, pessimism, suicidal ideation, and self-condemnation than those with PD without depression [13], [14]. The length of PD, Hoehn and Yahr (H&Y) phases, age, daily living, weak cognitive function, and sleep behavior disturbances are all linked to depression in PD, according to the research [15], [18].

Depression is common in the early stages of PD [19]. A previous study revealed that a lack of dopamine in Parkinson's disease causes sadness and dyskinesia [20]. Other studies, on the other hand, found that administering L-dopa, a Parkinson's therapy did not significantly alleviate depression symptoms [21]. As a result, it is impossible to say if depression is caused only because of a lack of dopamine. Many new studies have proposed that depression in Parkinson's patients is produced by a complex interaction of numerous elements rather than a single risk factor [22]. Regression models have been used in the past to forecast a risk factor. Individual risk variables can be investigated using basic regression models. However, because each study employed different confounding features, the studies were restricted in defining a risk factor when evaluating numerous risk factors. Furthermore, regression analysis necessitates data that meets several assumptions, including normality and linearity, which are highly likely to be violated when dealing with illness data. Various machine learning (ml) models such as the support vector machine (SVM) and logistic regression have been used in recent studies to reveal many risk variables for diseases like depression using statistical classification methodologies [23].

In this study, we use multi-modality features to reveal the inter-correlation of depression risk factors contributing to PD. We propose a hybrid deep-learning model consisting of a variational autoencoder, a 3D CNN, and a Bidirectional Long

short-Term Memory (Bi-LSTM) network. Using time-series data from 1059 patients, the network predicts PD patients with or without depression and a level of depression as a multitasking process. To summarize, our research makes the following contributions:

- We propose an end-to-end machine learning pipeline based on time series data to predict PD. The pipeline provided the possibility to explore different approaches of (i) fusion of data modalities, (ii) improving the quality of data using different techniques for missing values handling, data balancing, feature selection, data normalization, etc., (iii) optimization of different machine learning models, (iv) mapping of time series data to non-time series data, and (v) providing XAI capabilities.
- We optimize the PD progression detection problem as two classification tasks with different complexity levels, i.e., three-class and four-class tasks according to the H&Y scale (discussed in Section III-A3). For each task, five medically relevant time series modalities are explored, including Subject Characteristics (SC), Bio-Samples (BS), Medication History (MH), Motor function (M), and Non-Motor (MN) function modalities. Further, different fusions of these modalities are examined to detect the most critical modalities and the specific features in each modality.
- Real-world medical data are normally imbalanced. To solve this issue, variants of the over-sampling techniques such as SVM SMOTE, Borderline SMOTE, ADASYN (Adaptive Synthetic), SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous), and SMOTEENN (Synthetic Minority Over-sampling Technique with Ensemble of Neighbors) are compared, and the optimal technique is selected by performing multiple experiments on different data modalities
- To provide trustworthy and medically acceptable decisions, the optimized ML models are extended to provide explainable decisions; two explainers based on LIME and SHAPASH, are implemented to offer insight explanations for each classifier's decisions. The resulting models provide accurate and explainable decisions
- We propose a multimodal multitask hybrid deep learning architecture for the prediction of PD with a degree of depression. Our approach uses time series data from the PPMI database to jointly predict the severity of PD and depression in a patient.
- We combine two significant modality types, motor and non-motor, with the demographic characteristics of the subjects. In addition, to improve the performance of the proposed model, we use a neuroimaging modality, i.e., magnetic resonance imaging (MRI). Noting that the motor and nonmotor modalities are time series, and the subject demographics and neuroimaging modality have a baseline record only.
- The proposed model is composed of three main modules: (1) a variational autoencoder that extracts

deep features from the early fused numerical data of the motor, nonmotor, and demographic modalities; (2) a 3D CNN that captures in-depth features from the neuroimaging modality; and (3) a Bi-LSTM that performs multitasking prediction based on the decision fusion of the deep representations extracted from time-series, demographics, and MRI data.

- The performance of the proposed multitasking framework is evaluated to predict single-task and multitask problems.
- We conducted three single-task experiments in which our proposed model optimized on a specific classification task. In addition, the model has been optimized for multitasking problems to predict two or three classification tasks simultaneously.
- Each experiment has been analyzed with five distinct model configurations: (1) training the model with all of the numerical features, (2) training the model with all of the data's numerical features and with the Bayesian Optimizer (OPT), (3) training the model using the numerical features selected by the SOLUV feature recursive approach and with the optimizer, (4) training the model using the numerical features extracted by the variational autoencoder and with the optimizer, and (5) integrating the neuroimaging modality features from 3D CNN to train the model with the numerical features derived by the variational autoencoder and with the optimizer.
- The experimental results show that the proposed model is more medically perceptive, robust, and efficient than previous cutting-edge research.

The rest of the paper follows as outlined below: Section II discusses the related work in terms of PD and depression diagnosis and progression. Section III highlights the different PD and depression scales to predict the disease. In addition, a theoretical concept on variational autoencoders and Bi-LSTM is discussed. Section IV is the proposed architecture and discusses every component of the proposed framework. Section V discusses in-depth the experimental analysis along with the discussion and comparison with recent literature. Section VI discusses limitations and future challenges, and Section VII concludes the paper.

II. RELATED WORK

In clinical settings, depressive disorders are usually unrecognized and, even when found, generally under-treated [24]. To bridge that gap, this study focuses on the current understanding of the epidemiology, etiology, clinical characteristics, and depressive disorders in PD using deep machine learning. Recent developments in machine learning (ML) and deep learning (DL) have greatly improved the predictive modeling of Parkinson's disease (PD), hence guiding more holistic and data-driven clinical research. For feature-based classification, conventional ML techniques like Support Vector Machines (SVM) and Random Forest (RF) have been extensively applied [25], so efficiently detect important risk

variables connected with PD development. These techniques, however, can depend on manual feature engineering and might find it difficult to identify temporal relationships in longitudinal patient data. Deep learning models—including Long Short-Term Memory (LSTM) networks—have been progressively embraced to overcome these constraints because they capture sequential patterns and learn temporal dependencies from multimodal time-series data. Leveraging deep feature extraction and time-aware classification, hybrid architectures integrating CNNs, LSTMs, and attention mechanisms have further enhanced PD prediction [26]. These developments allow better integration of motor, non-motor, and neuroimaging data, hence enhancing diagnostic accuracy, disease progression modeling, and individualized treatment strategies [27]. Stacking ensemble machines, which improve accuracy by combining two or more single machine learning models with a meta-model, may reduce the risk of bias associated with a single machine learning model. The SVM separates data into two groups and searches for the best border between them. The goals of this work were to create a model for predicting DPD using the SVM while taking into account sociodemographic characteristics, health behaviors, Parkinson's symptoms, sleep behavior problems, and neuropsychiatric indicators as predictors, as well as to offer baseline data for detecting DPD. For extracting and learning information from voice samples, DL algorithms are available in a variety of forms. A multitask learning strategy is combined with an LSTM-based autoencoder in this work. There is no need to fine-tune feature preprocessing when utilizing an autoencoder architecture since the network learns the most relevant characteristics required for categorization (feature extraction) [28]. Because depression and Parkinson's disease both can have time-dependent characteristics, applying a network with memory can capture such information [29], [30], [31]. By using multi-task learning [32], the effect of overfitting can be avoided. In this study, besides the disease label, we use speaker IDs as target labels only, forcing the autoencoder to learn speaker-specific spectrum features. We hope to generate more robust features for disease classification. These findings highlight the need for early detection of depression in order for doctors to mitigate any detrimental effects on patients and carers, whether through antidepressant medication or extra support options [33]. Younger age, earlier onset of symptoms, female gender, increasing severity of motor symptoms, autonomic symptoms, olfactory deficiency, and a history of psychiatric symptoms have all been linked to depression in people with Parkinson's disease [34], [35], but these findings are not consistent across studies. Poorer sleep quality and impairment in everyday activities have also been linked to depression in cross-sectional studies. [36], there is still a lot of debate over the clinical correlates of depression in Parkinson's disease, as well as research into risk factors that potentially predict the future.

The loss of dopaminergic neurons in the substantia nigra and basal ganglia causes Parkinson's disease. Tremor,

postural instability, muscular stiffness, and slowness of movement are four of the most common symptoms. PD subtypes are not well defined; however, they are classified according to their severity levels. The degrees of PD progression are characterized as mild PD, moderate PD, and advanced PD [37], [38]. Widely used clinical instruments for evaluating PD severity include the Hoehn and Yahr (H&Y) scale and the Unified Parkinson's Disease Rating Scale (UPDRS), where H&Y offers a stage-based classification and UPDRS offers a comprehensible evaluation of motor and non-motor symptoms [39].

PD risk is increased by variations in the genetic mix of certain genes. According to studies, mutations in the LRRK2 (leucine-rich repeat kinase 2), PARK7 (Parkinsonism Associated Deglycase) [40], PRKN (Parkin RBR E3 Ubiquitin Protein Ligase), PINK1 (PTEN-induced putative kinase 1), or SNCA (alpha-synuclein) genes increase the risk of PD [41], [42]. GBA (glucocerebrosidase) and UCHL1 (Ubiquitin C-Terminal Hydrolase L1) are thought to affect the risk of PD development in some families. Individuals with GBA-related PD show a variety of phenotypes with an early age start and greater cognitive impairments than those without the disease, according to Sidransky and Lopez. They believe that understanding the genetics, pathogenesis, early intervention, and therapy of GBA-related PD can be improved by identifying pathogenic processes hidden in the illness [43]. The advancements and cell biological processes in GBA-associated PD recently [44]. GBA mutations have an impact on enzymatic functions as well as cognitive decline, and a decrease in GBA activity raises the risk of Alzheimer's disease. The primary neuropathological feature of Parkinson's disease is the increase of alpha-synuclein levels. In general, Research on Parkinson's disease (PD) has shown several modifiable risk factors that emphasize the intricate interaction between environmental exposures and genetic inclination. Increased PD risk has been associated to pesticides, herbicides, solvents, and heavy metals (e.g., manganese, lead); long-term occupational or residential exposure is a major contributing factor.

III. METHOD

In this section, we explain the most common scale used in predicting PD and the level of depression. Motor and non-motor function assessments (MDS-UPDRS Part III, sleep, and autonomic dysfunction scores) offer critical diagnostic signs, and the most widely used scales are the Hoehn and Yahr (H&Y) scale and the Unified Parkinson's Disease Rating Scale (UPDRS). Using the Geriatric Depression Scale (GDS) and MDS-UPDRS Part I, depression is evaluated. While a Bidirectional Long Short-Term Memory (Bi-LSTM) network captures temporal relationships for multitask categorization, a variational autoencoder (VAE) extracts latent representations from time-series data. Later, we will discuss the primary approaches used in our proposed architecture section, such as BiLSTM and VAE algorithms, which were tuned to act as classifiers for detecting PD progression and depression. In PD progress and depression

prediction, this combination improves accuracy and clinical relevance.

A. PD AND DEPRESSION PREDICTION SCALES

For PD, there are various measures used to assess patient impairment and disability i.e., stage and severity. The primary scales are established using common symptoms and biomarkers to place the patients in the right stage. The most widely used scales are the Hoehn and Yahr (H&Y) scale and the Unified Parkinson's Disease Rating Scale (UPDRS) [45].

The following is a concise description of the H&Y scales and other depression scales

1) HOEHN AND YAHR SCALE

The Parkinson's Hoehn and Yahr scale (H&Y) is a regularly used tool for determining the overall severity of the disease. The H&Y scale has a total of 5 stages, progress from unilateral (Stage 1) to bilateral without balance issues (Stage 2), followed by postural instabilities (Stage 3), loss of physical dependence (Stage 4), and finally confined to a wheelchair or bed until supported (Stage 5) [46]. When dividing Parkinson's disease into four stages, the healthy class (stage 1), the early class (stage 2), the advanced-early class (stage 3), and the advanced class (stage 4) were taken into account. Because there are few patients at stage 5, we omitted the stages beyond the fourth class. According to studies, these stages have a significant link with the indicators of standard of living and a strong association with the unified Parkinson's Disease Rating Scale (UPDRS) [46].

2) MOTOR AND NON-MOTOR FUNCTIONS

Motor function was the primary model result of interest. The MDS-UPDRS Part III score served as the main result. Demographics, illness duration, and motor (off-state) and non-motor function measurements were all used as predictive factors. Motor evaluations included the MDS-UPDRS Parts II-IV, PD subtype evaluations (tremor dominant, postural instability, and gait difficulty), To assess a patient's sleep quality and disruptions, the Epworth Sleepiness Scale (ESS), and the REM Sleep Behavior Disorder Screening Questionnaire Score (RBDSQ) were used. PD impulse control abnormalities were assessed using the QUIP (Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease) score. The Geriatric Depression Scale (GDS) score was created particularly for geriatric individuals to assess for depression. Autonomic nerve function was assessed in the previous month using the total score of the evaluation of autonomic dysfunction in Parkinson's disease (SCOPA-AUT). To assess mood and mentation, the MDS-UPDRS Part I score and component scores were employed [47].

3) MEASURES FOR DEPRESSION

The Geriatric Depression Scale (GDS) score was created particularly for geriatric individuals. The most appropriate measures for PD-related depression prediction among PD motor symptoms and conventional depression markers are

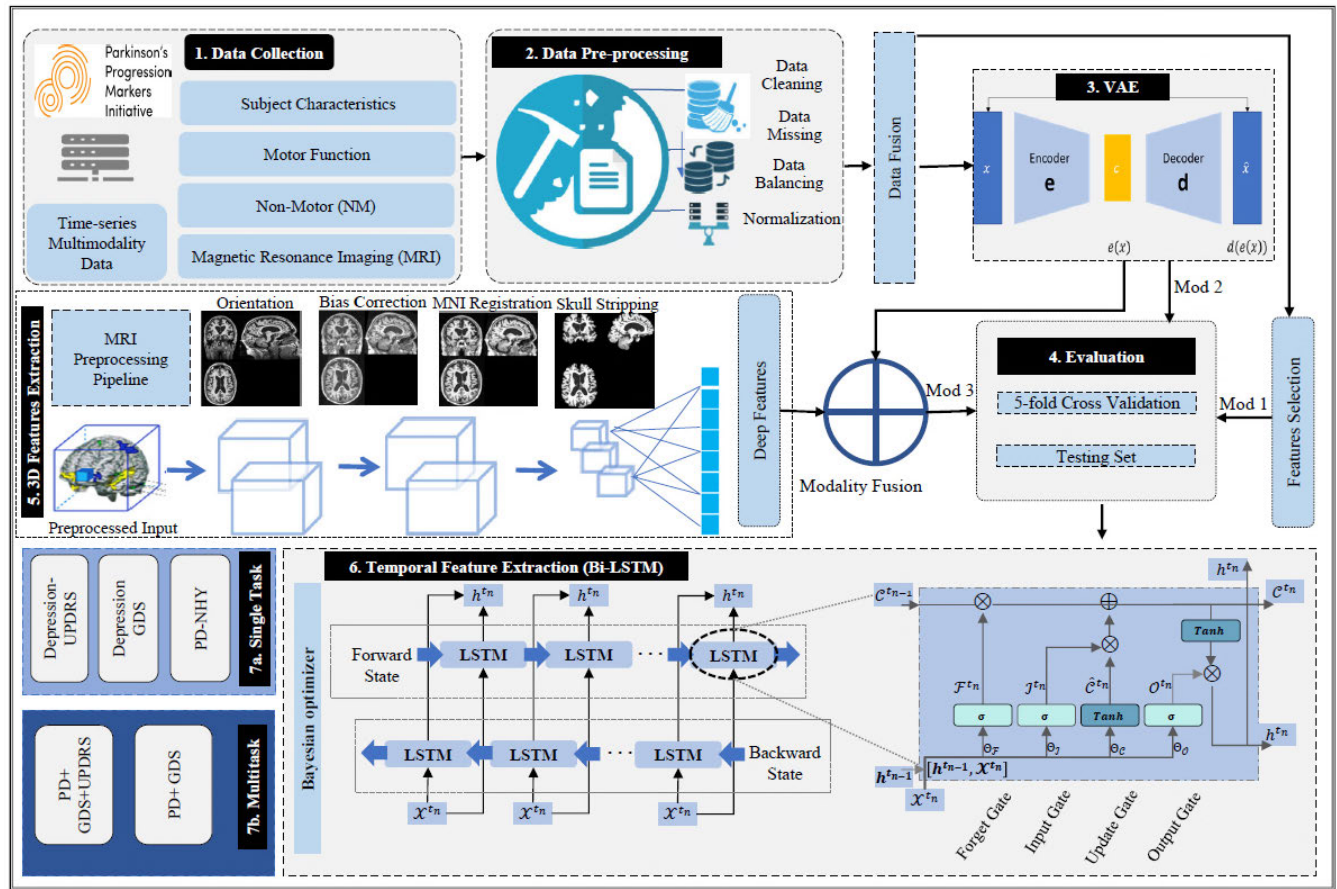


FIGURE 1. Detail of the proposed framework for the multitask classification. First, collect the dataset with multi-modalities. Second, several preprocessing steps are performed for the model stability, such as normalization and the class imbalance technique. Third, we used a variational autoencoder (VAE) for the feature extraction. Fourth, a 3D CNN extracts the MRI deep features and uses them to fuse VAE-obtained features. 5-fold cross-validation and testing are used for evaluation. The model is trained on the selected features and fused features. Ultimately, the results are evaluated on the testing set after hyperparameter optimization for single and multi-tasks. Mods 1, 2, and 3 are the approaches of experiments, explained in detail in the Results and Discussion section.

the Geriatric Depression Scale (GDS and MDS-UPDRS Part I scales. Because GDS excludes physical symptoms, it is recommended for self-reported screening; MDS-UPDRS Part I offers clinician-rated insight into mood and mentation. The MDS-UPDRS Part I score and component scores were employed. For discriminant and convergent validity, single items from the Movement Disorder Society—Unified Parkinson's Disease Rating Scale (MDSUPDRS) 27 were utilized. The items on this scale are graded by doctors and range from higher scores indicating more symptoms. Item 3 evaluates depressed symptoms, item 4 evaluates anxiety symptoms, and item 5 evaluates apathy symptoms in Part I. While these single items give minimal information about the prevalence of sadness or apathy, the Movement Disorder Society recommends using them to screen for either illness (especially item 5). Depression scales offer higher scores that point to worse depression; threshold values define a patient's level of non-depression, mild, moderate, or severe depression. Other measures, such as HAM-D and BDI, can still be helpful but need careful interpretation to differentiate PD-related physical disabilities from depressed symptoms [48].

B. VARIATIONAL AUTOENCODER

In 2013, the variational auto-encoder was first presented, and it is now frequently utilized in machine learning applications. Because the VAE is both probabilistic and generative, it differs from typical autoencoders. VAEs are autoencoders that address the problem of latent space irregularity by having the encoder return a distribution over the latent space rather than a single point, and by including a regularization term over that returned distribution in the loss function to ensure better latent space organization. Traditional autoencoders aim to learn a compressed representation of input data but do not impose a probabilistic structure on the latent space, making it difficult to control the learned representations effectively. This limitation is addressed by Variational Autoencoders (VAEs), which introduce a probabilistic perspective by modeling the latent space with a prior distribution, typically a multivariate Gaussian $z \sim \mathcal{N}(0, 1)$. Using the statistical approach of variational inference (thus the name), the rather obvious loss function of VAEs, comprising of a reconstruction term and a regularization term, can be carefully constructed, given a basic underlying probabilistic model to represent

our data [49], [50], [51]. VAE is a fully connected neural network-based unsupervised learning approach. The encoder and decoder layers have multilayer perceptron structures, and the node computation process is similar to that of a fully connected neural network. Through the encoder layer structure, VAE translates the data information in the samples to the low-dimensional latent space [52], [53].

Following underlying data distribution patterns, the decoder layer rebuilds the samples by decoding the information in the latent space. The data variable $\mathcal{X}$ is mapped onto the latent space $\mathcal{Z}$, with the latent variable z following the usual Gaussian distribution $z \sim \mathcal{N}(0, 1)$. Assuming that z has the distribution function $\mathcal{G}(z)$, the data $\mathcal{X}$ may be created using the function defined below:

$$\mathcal{G}_\Theta(\mathcal{X}) = \int \mathcal{G}_\Theta(z) \mathcal{G}_\Theta(\mathcal{X} | z) dz \quad (1)$$

where Θ is the model parameter that was created. The probability function of $\mathcal{G}_\Theta(\mathcal{X})$ could be represented as follows using the concept of variational inference:

$$\log \mathcal{G}_\Theta(\mathcal{X}) = \mathcal{KL}(q_\phi(z | \mathcal{X}) \| \mathcal{G}_\Theta(z | \mathcal{X})) + \Gamma(\Theta, \phi; \mathcal{X}) \quad (2)$$

where Kullback-Leibler divergence is termed as $\mathcal{KL}$ in the above equation, and variational lower bound denoted by $\Gamma(\cdot)$. The optimization of the above equation can be further expressed as:

$$\Gamma(\Theta, \phi; \mathcal{X}) = -\mathcal{KL}(q_\phi(z | \mathcal{X}) \| \mathcal{G}_\Theta(z)) + \varepsilon_{q_\phi(z | \mathcal{X})} [\log(\mathcal{G}_\Theta(\mathcal{X} | z))] \quad (3)$$

Our goal is to $\Gamma(\cdot)$ if it has to be optimized can be expressed as:

$$\Gamma(\Theta, \phi; \mathcal{X}) = -\mathcal{KL}(q_\phi(z | \mathcal{X}) \| \mathcal{G}_\Theta(z)) + \log(\mathcal{G}_\Theta(\mathcal{X} | z)) \quad (4)$$

Our goal is to $\Gamma(\cdot)$ if it has to be optimized can be expressed as:

$$\Gamma(\Theta, \phi; \mathcal{X}) = -\mathcal{KL}(q_\phi(z | \mathcal{X}) \| \mathcal{G}_\Theta(z)) + \log(\mathcal{G}_\Theta(\mathcal{X} | z))$$

The objective function may be represented after resetting the starting values of several distributions and deleting the constant term:

$$\{\Theta, \phi\} = \arg \min_{\Theta, \phi} \left(-\frac{1}{2} \sum_{d=1}^{D_l} \left(1 + \log \sigma_{zd}^2 - \mu_{zd}^2 - \sigma_{zd}^2 \right) + \frac{1}{2} \sum_{d=1}^{D_y} (x_d - \mu_{xd})^2 \right) \quad (5)$$

After resetting the beginning values of various distributions and removing the constant term, the objective function may be written as:

$$\mathcal{F} = \alpha \left(\arg \min_{\Theta, \phi} \left(-\frac{1}{2} \sum_{d=1}^{D_l} \left(1 + \log \sigma_{zd}^2 - \mu_{zd}^2 - \sigma_{zd}^2 \right) \right. \right.$$

$$\left. + \frac{1}{2} \sum_{d=1}^{D_y} (x_d - \mu_{xd})^2 \right) + \beta (x - x')^2 \quad (6)$$

where α and β are the same in the range of (0, 1), corresponding to the weights of two optimization targets. The error function of VAE is $\mathcal{F}$, which modifies the weights and biases between the encoder and decoder layers repeatedly. In reality, $\mathcal{G}_\phi(\cdot)$ is the process of mapping $\mathcal{X}$ to the latent space $\mathcal{Z}$ and producing the mean μ and variance σ^2 using the encoder. Similarly, $q_\phi(\cdot)$ is the process of reconstructing $\mathcal{X}$ from a latent variable z using the decoder layer [54], [55], [56]

C. BIDIRECTIONAL LONG SHORT-TERM MEMORY

This study chose Bidirectional Long Short-Term Memory (Bi-LSTM) because of its efficiency in capturing long-term dependencies in time-series data, robust handling of sequential patterns, and proven effectiveness in medical applications related to disease progression modeling. Unlike typical recurrent networks, LSTM successfully mitigates the vanishing gradient problem, making it well-suited for modeling longitudinal data patterns in Parkinson's disease (PD) development. The bidirectional mechanism improves performance by learning from both past and future settings, which is critical for increasing diagnostic accuracy in sequential clinical examinations [57]. Compared to other deep learning-based forecasting approaches such as N-BEATS [58] and DeepAR [59], Bi-LSTM provides better interpretability and adaptability for irregular time-series data, which is frequent in clinical datasets. While N-BEATS excels in time-series forecasting by leveraging deep residual networks, it largely focuses on point predictions and does not clearly express temporal relationships as well as LSTMs. Similarly, DeepAR, an autoregressive model built for probabilistic forecasting, requires a huge number of identical time-series data, making it less appropriate for a heterogeneous clinical dataset such as PPMI. In studies it is shown that the LSTM-based architectures outperform N-BEATS and DeepAR in medical time-series prediction tasks because of their capacity to describe complicated sequential relationships and illness progression dynamics [60], [61]. Furthermore, Bi-LSTM exhibits a fair trade-off between accuracy, computational efficiency, and generalizability, making it appropriate for real-world clinical applications.

IV. PROPOSED ARCHITECTURE

The proposed multimodal multitask deep learning framework outperforms existing PD progression and depression detection models by integrating motor, non-motor, and MRI-based features. Unlike prior studies that rely on single-task classification or unimodal data, our model fuses Variational Autoencoder (VAE) for feature selection, 3D CNN for MRI-based deep feature extraction, and Bi-LSTM

for time-series modeling, achieving high accuracy for NHY classification. The multitask approach improves generalization, predicting PD severity, depression presence, and severity with increasing accuracy over 2-time step, 4-time steps, and 6-time steps. Bayesian Optimization and SMOTE-based balancing further enhance stability and efficiency. By combining deep feature extraction, temporal modeling, and multitask learning, the proposed method surpasses traditional and existing deep learning approaches, offering a robust and clinically relevant PD progression prediction model.

The important characteristics using multimodal time-series data encompass demographic, motor, non-motor, and neuroimaging (MRI) aspects. While motor function measurements, including the MDS-UPDRS Part III score, bradykinesia, tremor severity, and postural instability scores, demographic characteristics such as age, gender, and illness duration give crucial background information. The Geriatric Depression Scale (GDS), sleep disorder indices (Epworth Sleepiness Scale, REM Sleep Behavior Disorder score), and autonomic dysfunction scores (SCOPA-AUT) among non-motor characteristics help to evaluate the neuropsychiatric effect of the disease. Furthermore, 3D CNN processing of neuroimaging data from MRI scans extracts deep spatial patterns reflecting structural brain abnormalities connected with Parkinson's disease. Via a hybrid technique, these retrieved characteristics are combined into a deep learning pipeline. Whereas a 3D CNN generates deep representations from MRI scans, a variational autoencoder (VAE) is used for feature extraction and dimensionality reduction from the numerical time-series data. Three classification tasks—(1) NHY stage classification for PD severity, (2) binary classification of depression present, and (3) four-class classification of depression severity are then predicted by a multitask Bidirectional LSTM (Bi-LSTM) network fed the fused features from VAE and 3D CNN. Hyperparameter tuning uses Bayesian optimization, a 5-fold cross-validation method that guarantees the strength of the suggested model. Deep learning methods and multimodal data integration improve prediction accuracy, therefore allowing a more complete and clinically meaningful evaluation of PD and depression development.

In our proposed work, algorithmic validation, anomaly detection, and optimization methods are used at several phases to find and fix errors. By use of performance assessment measures like accuracy, precision, recall, and F1-score—which serve to quantify misclassifications—the deep learning model detects mistakes and errors. By identifying overfit or underfit, cross-validation methods—including 5-fold cross-validation ensure model stability. Data preprocessing techniques, including median filling for continuous data and mode filling for categorical variables, handle anomalies in multimodal time-series data—that is, missing or inconsistent values. Using feature selection methods such as the Variational Autoencoder (VAE) and the SULOV algorithm, the suggested study reduces duplicate and point-less features, hence lowering classification errors resulting

from noisy data. Hyperparameter tuning is done via Bayesian optimization to guarantee the best model performance and lower errors. By learning long-term trends in time-series data, Bi-LSTM also efficiently manages temporal dependencies, hence reducing mistakes in NHY progression prediction. Moreover, adding extra information on brain structural changes, including MRI-derived deep features generated using 3D CNN improves classification accuracy. Confusion matrices help any misclassified instances be investigated for trends in mispredictions.

A. DATA COLLECTION

This research is conducted using the PPMI database (<http://www.ppmi-info.org/data>). The database is a multinational, large-scale database that records the progression of PD. The database is comprehensive in its coverage of patient clinical information. The PPMI dataset includes medical images, clinical reports, motor, and non-motor test results, biosamples, and each patient's history and background. The inquiry extracts a range of data classes from the dataset, each with its own set of features [63]. The study covers the essential modalities, such as Subject characteristics, Motor Function, and Non-Motor function. The dataset for PD is in the form of time series (as patient visits) from the PPMI data. For each patient, six common visits are sorted within every 12-month interval, i.e., baseline (BL) in month 0, visit 1 in month 12, visit 2 in month 24, visit 3 in month 36, visit 4 in month 48, and visit 5 in month 60 as shown in Figure 2. Only non-time series data are accessible in the SC modality, although all other modalities offer time series observation. Each patient's class is determined using the variable NHY based on the H&Y scale. The datasets are prepared, including for the four-class problem ($n=1059$) and for MRI ($n=840$).

For generalization purposes, the amount of data is critical in ensuring trustworthy and reproducible findings. The combination of 5-fold cross-validation and SMOTE-based class balancing enables robust training and prevents overfitting, while the Bi-LSTM + 3D CNN fusion model captures illness development accurately. Compared to smaller datasets (<500 participants), this dataset gives a more thorough representation, making it reliable for predicting Parkinson's disease severity and depression levels. External validation can further increase dependability, although the dataset is big and varied enough to assure robust generalization.

B. DATA PRE-PROCESSING

We performed two preprocessing pipelines: (a) for numerical features modalities including Subject Characteristics (SC), Motor function (M), and Non-Motor (MN) function modalities, and (b) MRI. The preprocessing of numerical modalities is discussed in this subsection. For the MRI preprocessing details, can be found in section IV-D.

1) DATA PREPARATION

Before training the proposed model, multiple data preparation steps are performed. The first step is to identify the most

effective modalities for detecting PD. The discriminatory capability of each distinct modality for detecting PD severity and most modalities' statistical significance is evaluated. The heterogeneous time series data is subjected to a set of preprocessing processes to enhance the data quality. The following are some of the steps in the process: Missing data for all modalities is handled in the first stage. All features with a missing data percentage exceeding 30 percent are eliminated. The missing data are handled in various ways, including forward and backward filling [64]. The median filling is used for the continuous (numerical) features, and the mode technique is used to fill categorical data. While mode imputation maintains categorical structure without creating false patterns, median imputation reduces absolute deviation and is less susceptible to skewed distributions [95], [96].

2) DATA SPLITTING

Our dataset is comprehensive compared to the literature studies. In addition, our models are based on deep learning models. We utilize an accurate method to divide the dataset into training and validation. The 5-fold cross-validation technique was used with the 70% partition to train and optimize the proposed deep learning model using 30% of the validation set and the average results with standard deviations are reported.

3) DATA BALANCING

the dataset we have is unbalanced. Initially, we apply SMOTE, SMOTE+ENC [65], and other balancing techniques that generate fake data, and we use these data to balance our dataset. Although the resulting dataset achieved high performance, generating fake data for data balancing is not recommended in the medical field, because physicians will not trust the results. Another method for data balancing is called undersampling. The balance is achieved by lowering the size of the plentiful class at random while retaining all samples in the uncommon class. This strategy, however, wastes a great deal of data. As a result, scientists used oversampling techniques to their advantage. Therefore, we apply the oversampling technique [66] separately for training and validation datasets. This technique balances the dataset by increasing the size of rare samples by adding new samples from the rare class (for example, by repetition, bootstrapping, or a synthetic minority).

4) DATA SCALING

We employed a min-max normalization approach of the created statistical features before feeding it to the deep-learning model in order for the model to converge rapidly and decrease training time. The associations between the original data values are preserved by min-max normalization. We will have reduced standard deviations as a result of this restricted range, which can reduce the impact of outliers. We use the following formula to convert each feature [67].

$$f(i, j) = \frac{f(i, j) - K_{\min}}{K_{\max} - K_{\min}} \quad (7)$$

where $K_{\min}$ and $K_{\max}$ are the minimum and maximum values in the picture f , and $f(i, j)$ represents each pixel in the brain image.

5) FEATURE SELECTION

In most machine learning modeling experiments, selecting appropriate input characteristics for result prediction is a typical challenge [68]. The fundamental concept is to find the minimal amount of input characteristics that can nevertheless produce decent predicted results. Furthermore, lowering the feature dimension reduces the likelihood of over-fitting and eliminates uninformative features, allowing the classifier to focus on informative characteristics [69]. For this purpose, the best feature selection technique is used after experimenting with multiple techniques. Recursive XGBoost and Searching for Uncorrelated List of Variables, or SULOV which is based on the Minimum-Redundancy-Maximum-Relevance (MRMR) algorithm, one of the greatest feature selection techniques, is the foundation of the SULOV algorithm. is used to reduce features in order to select the best features for the model [70].

Consider a data set $D = \{(\mathbf{X}_i, \mathbf{Y}_i) \mid i = 1, \dots, n, \mathbf{X}_i \in \mathbb{R}^m, \mathbf{Y}_i \in \mathbb{R}^m\}$, with number of samples n and features m . Let $\hat{\mathbf{Y}}_i$ be computed:

$$\hat{\mathbf{Y}}_i = \sum_{i=1}^n \mathcal{T}_i(\mathbf{X}_i) \quad (8)$$

where a regression tree is $\mathcal{T}_i$. An objective function is minimized to learn $\mathcal{T}_i$ as follows:

$$O = \sum_{i=1}^n L(\mathbf{Y}_i, \hat{\mathbf{Y}}_i) + \sum_{i=1}^n \Omega(\mathcal{T}_i) \quad (9)$$

By means of a feature selection procedure, the key characteristics for Parkinson's disease (PD) were chosen including both motor and non-motor functions modalities. Six visits of time-series data allowed the chosen top features, based on their severity ratings using the Hoehn and Yahr (H&Y) scale—to be retrieved. A deep learning-based prediction model runs on these inputs. Early fusing of these multimodal characteristics, followed by feature extraction using a Variational Autoencoder (VAE), is how approaches section A, B, and C integrate as detailed in Section III. A 3D CNN also generates deep representations from MRI data, which are then combined with VAE-derived features. A Bidirectional LSTM (Bi-LSTM) model processes these fused features allowing for multitask categorization. This method distinguishes between depressed and non-depressed individuals, forecasts PD intensity on the NHY scale, and ranks depression degree. With matching feature codes accessible in the Supplement file attached, In second file S1-Fig-A graphically shows the chosen characteristics depending on their severity classifications. This approach uses deep learning to incorporate temporal and multimodal relationships, hence improving prediction accuracy and producing a strong and complete PD progression model.

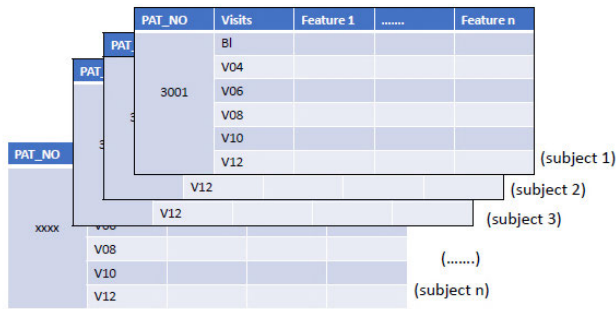


FIGURE 2. Describe the structure of the time series dataset. Each patient has six visits including the baseline. The patient's demographic, motor, and non-motor modalities are time series data. The MRI modality is non-time series data.

C. FEATURE EXTRACTION VIA VARIATIONAL AUTO-ENCODER

In VAE, the encoder projects the input data to a stochastic distribution of the latent variable, the latent variable is then sampled from the distribution, and the decoder reconstructs the input data using the latent variable. To put it another way, a VAE network that has been trained to properly recreate training data also suggests that the latent variable contains enough input data information. Furthermore, because the latent variable's dimension is usually less than that of the input data, the encoder can generate a low-dimension representation of the data. To put it another way, VAE may help with data processing tasks like feature extraction. Previously published work [54], In classification tasks, VAE has been shown to have strong feature representation capabilities. Because of the reconstructive training aim of VAE as a new non-linear feature representation approach, features obtained by VAE may not only retain critical data information but also eliminate irrelevant and superfluous information. When compared to AE, VAE characteristics are expected to include more information since it has a probabilistic structure rather than a defined structure [71], [72], [73]. Some common features are extracted from the development of movement disorders in supplement file S2- fig-A, shows the most common extracted features, while list of full features are in supplement S1 file. the obtained characteristics comprise motor-related markers including Unified Parkinson's Disease Rating Scale (UPDRS) scores, tremor intensity, gait instability, and bradykinesia severity. Depression severity (Geriatric Depression Scale), sleep disorder scores (Epworth Sleepiness Scale, REM Sleep Behavior Disorder Screening), and autonomic dysfunction scores (SCOPA-AUT) are among non-motor factors included to explain psychiatric and cognitive decline in Parkinson's patients. Furthermore, adding context to improve prediction accuracy are demographic variables like age, gender, and illness duration.

D. MRI MODALITY AT BASELINE

Magnetic resonance imaging analysis is a valuable tool for acquiring, preprocessing, extracting useful features,

selecting, and classifying MRI pictures, among other things. Element abstraction is a method of portraying a picture with raw data by performing processing to extract useable data from the image in order to improve decision-making processes such as pattern classification. The goal of characteristic extraction is to limit the amount of unique data by deciding on certain assets, or characteristics, that distinguish one data framework from another. Towards detecting those representations of the image's significant characteristics that are based on characteristic vectors. Sorts, the properties of substances vying for attention, albeit these should be chosen with caution, and the portion should be transformed into a delegate test of the most important [74], [75].

1) MRI PRE-PROCESSING

Image pre-processing is the process of removing unnecessary information from the RAW MRI volumes, allowing for easy comparison of various brain scans. In comparison to using raw MRI volumes as the input data, we noted a significant improvement in the model's performance after performing this step. The pre-processing pipeline employed for this study includes image reorientation, bias field correction, skull stripping, and registration of MNI template space. In this study, we used 800 magnetic resonance imaging volumes available at the baseline for each patient. Our dataset is composed of 3T T1-weighted anatomical sequences were recorded utilizing the volumetric 3D MPRAGE protocol with a resolution of a $1 \times 1 \times 1$ mm, voxel size. All MRI volumes were passed through a standard pre-processing pipeline, as shown in the pre-processing section of Figure 1 step 4. After the pre-processing step, 110 most informative coronal slices are then extracted from the preprocessed input volumes. Slice selection is made after calculating the total half-volume size measured on either side of the central slice in the 3D volume. In this way, we hope to use only the most informative 2D slices, this enables the proposed network to capture the most representative feature vectors for PD.

2) REORIENTATION TO STANDARD SPACE

Sometimes MRI volumes are flipped by 180 degrees during the formation time at the MRI scanner. It's very crucial to reorient the flipped sides of MRI volumes to their accurate state. These operations are the basis of this essential pre-processing step that allows us to maintain consistency in voxel processing and interpretation across a variety of software and systems. Visual inspection of MRI volumes is the best practice to inspect the data before performing any further processing. In this study, we used FSLEye Obtainable from (<https://git.fmrib.ox.ac.uk/fsl/fsleyes/fsleyes>) the FSLEyes viewer module to visualize our data and the "fslre-oriented2std" tool to realign 180-degree flipped volumes in our data [76].

3) BIAS FIELD CORRECTION

Many factors cause noisy signals in the formation of MRI volume, and thus cause brightness issues in the resulting

TABLE 1. CNN architecture.

Layer ID	Layer Name	Number of Kernels	Kernel Size/Stride	Output Size
0	Input	-	-	$110 \times 110 \times 110$
1	Conv1	8	$3 \times 3 \times 3/1$	$8 \times 110 \times 110 \times 110$
2	Conv2	8	$3 \times 3 \times 3/1$	$8 \times 110 \times 110 \times 110$
3	MaxPool3D	-	$2 \times 2 \times 2/2$	$8 \times 55 \times 55 \times 55$
4	Dropout (0.3)	-	-	$8 \times 55 \times 55 \times 55$
5	Conv3	16	$3 \times 3 \times 3/1$	$16 \times 55 \times 55 \times 55$
6	Conv4	16	$3 \times 3 \times 3/1$	$16 \times 55 \times 55 \times 55$
7	MaxPool3D	-	$2 \times 2 \times 2/2$	$16 \times 27 \times 27 \times 27$
8	Dropout (0.3)	-	-	$16 \times 27 \times 27 \times 27$
9	Conv5	32	$3 \times 3 \times 3/1$	$32 \times 27 \times 27 \times 27$
10	Conv6	32	$3 \times 3 \times 3/1$	$32 \times 27 \times 27 \times 27$
11	Conv7	32	$3 \times 3 \times 3/1$	$32 \times 27 \times 27 \times 27$
12	MaxPool3D	-	$2 \times 2 \times 2/2$	$32 \times 13 \times 13 \times 13$
13	Dropout (0.4)	-	-	$32 \times 13 \times 13 \times 13$
14	Conv8	64	$3 \times 3 \times 3/1$	$64 \times 13 \times 13 \times 13$
15	Conv9	64	$3 \times 3 \times 3/1$	$64 \times 13 \times 13 \times 13$
16	Conv10	64	$3 \times 3 \times 3/1$	$64 \times 13 \times 13 \times 13$
17	MaxPool3D	-	$2 \times 2 \times 2/2$	$64 \times 6 \times 6 \times 6$
18	Dropout (0.4)	-	-	$64 \times 6 \times 6 \times 6$
19	Flatten layer	-	-	13824

MR slices. The most common reason for such a problem is the patient's position in the MR scanner or the version of the scanner itself. Such brightness issues in MR images are caused by low frequency, smoothed, and undesirable signals inside the scanner, this problem can affect the overall quality of all slices. The efficacy of the subsequent process, such as the tissue segmentation and skull stripping might be significantly impacted if such inhomogeneities are not corrected. In our scenario, the N4 bias field correction algorithm from Advanced Normalization Tools [77] (ANTs) was utilized, this is an advanced N3 bias field correction algorithm that uses an improved B-spline fitting routine to allow multiple resolutions in the correction process.

4) SKULL STRIPPING

In an MRI, skull stripping is the process of separating brain tissues from non-brain tissues. The dura is the outermost region of a skull made of a thick membrane from densely connective tissues. One of the main functionalities of dura matter is proving to protective envelope to the human brain. In the deep learning-based In MRI processing, dura regions are considered extra information and act as noise and contribute to the high dimensionality of the training data while making the classification task more complex. This is a crucial step to reduce the amount of unnecessary information. We used a software program called the Brain Extraction Tool in this study (BET2) [78], was implemented in FSL to remove dura and others.

5) MNI TEMPLATE SPACE REGISTRATION

Template registration is the process of aligning images based on brain structures to simplify the process of comparing different images. In this paper, the MNI-152 [79] template space was utilized to carry out affine transformation. This process does not deform the image because it only involves basic transformation steps such as rescalings, rotations, translations, and shearing. FMRIB's Linear Image Registration Tool (FLIRT) from FSL was used to perform the registration, and correlation ratios were used as the similarity metric.

6) DEEP FEATURES EXTRACTION USING 3D CNN

In our experiments, we used an efficient 3D CNN to extract the feature vector from the input volume at each time step. The proposed network uses an efficiently designed 3D CNN model to learn intra-slice features from the input volume. The input volume to the 3D CNN is of size $110 \times 110 \times 100$. The 3D CNN has 10 convolution layers and 4 max-pooling layers as shown in Table 1. The max-pooling layers are employed after the 2th, 4th, and 7th convolution layers. We did not use a max-pooling layer after each convolutional layer to preserve the spatial and temporal dimensions of the input volume to learn a wide range of features in both directions. The disadvantage of not using the max pooling layers after each convolutional layer is that training time can increase. However, this problem was addressed by finding the optimal number of kernels and kernel size at each layer in the proposed 3D CNN. We used 8 kernels of size $3 \times 3 \times 3$ in the 1st and 2nd layer, then the spatial and depth dimension of the feature maps with a $2 \times 2 \times 2$ max-pooling layer was reduced by stride 2 and followed by a 30% drop-out layer. We increased the number of kernels as the spatial dimensions of the feature maps decreased and the depth dimensions of the feature maps increased. 16 kernels are used at the 3th and 4th convolutional layers, the outputs of these layers go through the ReLU activation function, max pooling, and a 30% dropout layer. 32 kernels are used at the 5th, 6th, and 7th layers with the ReLU activation function, a max-pooling layer, and a 40% dropout layer. Finally, 64 kernels are used at the 8th, 9th, and 10th layers with ReLU activation function, max-pooling layers, and a 40% dropout layer. The detailed CNN

architecture parameters can be found in Table 1. Throughout the network, we used the ReLU activation function. The output of the last convolutional layer is flattened to a 1D feature vector that represents the input volumes from each of the time steps. The output feature vector from the VAE is fused with the deep features coming from the 3D CNN and is forwarded to the Bi-LSTM network to capture the temporal features using multimodal data of a patient.

E. REGULARIZATION STRATEGIES

To make our hybrid model architecture—which consists of the Variational Autoencoder (VAE), 3D Convolutional Neural Network (3D CNN), and Bidirectional LSTM (Bi-LSTM)—more robust and less prone to overfit, we employed a variety of additional regularization techniques in addition to standard dropout layers. To ensure that the model would function with new data and with various data splits, we first used 5-fold cross-validation with early stopping based on validation loss. Second, by placing a bottleneck in the latent space that was restricted to a Gaussian distribution, the VAE functioned as an implicit regularizer as well as a non-linear feature compressor. This reduces noise and excess data in the multimodal, high-dimensional frac. Importantly, the VAE was retrained on the training set in each fold independently, ensuring that latent features were generated without any test data exposure. This eliminated the possibility of data leaks. Previous studies have demonstrated that the variational aim, which balanced reconstruction error with Kullback-Leibler divergence, produced small and distinguishable representations that significantly enhanced classification performance [56]. Third, key hyperparameters such as the VAE latent dimensionality, the number of Bi-LSTM units, and the learning rates were adjusted using Bayesian Optimization. This prevented the model from becoming overly complex or over-parameterized. Fourth, to address the class imbalance without changing the validation results, we simply applied SMOTE-based oversampling to the training data. Finally, to penalize excessive weights and further restrict the model's capacity, an L2 regularization term was introduced into the Bi-LSTM loss function (15). These strategies combined to form a robust, multi-layered regularization framework that ensured the proposed model was stable, dependable, and applicable to a range of scenarios, despite its complex construction and moderately large dataset.

F. PROPOSED Bi-LSTM ARCHITECTURE

5-fold Cross-Validation Our Proposed LSTM network jointly learns the two correlated tasks:

- 1) y^1 : a four-class NHY scale that defines the stage of PD,
- 2) y^2 : a binary class that determines whether or not a patient is depressed. Both tasks are multi-class classification,
- 3) y^3 : a four-class depression scale that indicates the severity of depression.

The study depends on the diverse family of PD patients with depression and the level of depression of two

multivariate datasets represented as $S = \{\mathcal{D}^1, \mathcal{D}^2\}$. Each dataset, $\mathcal{D}^m = \{\mathbf{X}_1^m, \dots, \mathbf{X}_i^m, \dots, \mathbf{X}_N^m\}$ has N PD patients where each patient's $\mathbf{X}_i^m$ is a multivariate time-series sequence, $\mathbf{X}_i^m = \{\mathbf{X}_{i1}^m, \mathbf{X}_{i2}^m, \dots, \mathbf{X}_{if}^m\}$ for $t = (1, \dots, N)$ time-step for i -th dataset. As a result, every dataset is defined as $\mathcal{D}_i = \{\mathbf{X}_{i1}^m, \mathbf{X}_{i2}^m, \dots, \mathbf{X}_{if}^m, y_i^1, y_i^2\}$, $i = 1, \dots, N$, where y_i^1 is the discrete label of NHY stage, and y_i^2 is the class label of the depression for the i -th instance.

The optimization is performed on shared parameters (Θ^{sh}) simultaneously based on the specific task (Θ^t) whether NHY stage classification or depression level or depressed/non-depressed patients. The hypothesis of each task is parametrized as $f^t(\mathbf{X}, \Theta^{sh}, \Theta^t) : \mathbf{X} \rightarrow y$, and the specific loss functions of both classification tasks are $\mathcal{L}(\cdot) : y^t \times y^t \rightarrow \mathbb{R}^+$. The multi-objective optimization problem may be expressed to minimize the proposed model loss, as shown below:

$$\begin{aligned} \min_{\Theta^{sh}, \Theta^1, \Theta^2} \mathcal{L}(\Theta^{sh}, \Theta^1, \Theta^2) \\ = \min_{\Theta^{sh}, \Theta^1, \Theta^2} \left(\mathcal{L}^1(\Theta^{sh}, \Theta^1) + \mathcal{L}^2(\Theta^{sh}, \Theta^2) \right) \end{aligned} \quad (10)$$

where $\mathcal{L}^t(\cdot)$ is the loss function for a specific task represented as

$$\mathcal{L}^t(\Theta^{sh}, \Theta^t) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f^t(\mathbf{X}_i; \Theta^{sh}, \Theta^t), y_i) \quad (11)$$

We addressed both tasks in a common objective function defined as follows, for m is the number of Θ^{st} and Θ_t parameters.

$$\mathcal{L}(\Theta^{sh}, \Theta^t) = - \underbrace{\left[\frac{1}{N} \sum_{i=1}^N y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]}_{\text{Cross-Entropy Loss}} \quad (12)$$

$$+ \underbrace{\frac{\lambda}{m} \sum_{j=1}^m \Theta_j^2}_{\text{Regularization Term}} \quad (13)$$

where the cross-entropy loss is used for the multi-class classification task, with the term at the end being regularization.

$$p_i = \frac{1}{1 + \exp(-W^T \mathbf{X}_i)} \quad (14)$$

To regularize the Bi-LSTM network and reduce overfitting, we include an L_2 penalty term in the multi-task loss function. The revised objective becomes:

$$\begin{aligned} \mathcal{L}(\Theta_{sh}, \Theta_1, \Theta_2) = \sum_{t=1}^T \mathcal{L}_t(\Theta_{sh}, \Theta_t) \\ + \lambda \left(\|\Theta_{sh}\|_2^2 + \sum_{t=1}^T \|\Theta_t\|_2^2 \right) \end{aligned} \quad (15)$$

where $\mathcal{L}_t$ is the cross-entropy loss for task $t \in \{1, 2, 3\}$ (NHY stage, depression binary, and depression level), Θ_{sh} are

shared parameters, Θ_t are task-specific parameters, and λ is the regularization strength.

Each individual task loss is defined as:

$$\mathcal{L}_t(\Theta_{sh}, \Theta_t) = -\frac{1}{N} \sum_{i=1}^N y_i^{(t)} \log \hat{y}_i^{(t)} \quad (16)$$

The final objective balances task-specific accuracy and model simplicity, improving generalization on limited data.

G. BAYESIAN OPTIMIZATION

For a closed box function for which we lack a closed-form expression, we use the notation Φ . Additionally, evaluating this closed box function is costly. Let $\Phi : \mathcal{Z} \rightarrow \mathbb{R}$ be a properly behaving function that is defined on the subset $\mathcal{Z} \subseteq \mathbb{R}^d$. The following global optimization issue is what we're trying to resolve.

$$\mathbf{z}^* = \arg \max_{\mathbf{z} \in \mathcal{Z}} \Phi(\mathbf{z}). \quad (17)$$

By creating a probabilistic model for the closed box function $\Phi(x)$, Bayesian optimization seeks to identify the global optimum of the function while incorporating uncertainty. It then uses this model to decide where in $\mathcal{Z}$ to do the function's subsequent evaluations. This produces a method that finds the smallest number of challenging non-convex functions with just a small number of evaluations but at the cost of needing to do more calculations to know where to attempt next.

Combining fixed parameters with Bayesian optimization-based tuning in the proposed framework guarantees a balance between model stability and generalizing capacity the hyperparameter selection technique. To minimize needless computing overhead and prevent too great fluctuation in outcomes, several important hyperparameters were established depending on past empirical research and best practices in deep learning for neuroimaging analysis and medical time-series. Fixed hyperparameters comprise batch size (64), optimizer (Adam), activation function (ReLU), and dropout rates (30–40%), since these values routinely produce steady convergence and prevent overfitting in long-sequence time-series and CNN-based feature extraction. Critical hyperparameters including latent space, hidden units, and learning rate, however, are not excluded. Fixed and optimal hyperparameters together provide model stability, reduce overfitting, and enhance generalization performance over several degrees of PD intensity. While fixed parameters were selected according on past medical AI research [16], [17], the size of the variational autoencoder (VAE) was auto-tuned using Bayesian Optimization, thereby maximizing the trade-off between model complexity and prediction accuracy.

1) BAYESIAN INFORMATION FUSION

The fusion of diverse modalities, such as time-series clinical data and MRI imaging, lacks a well-established mathematical foundation for improving diagnostic certainty.

The discrepancy between time-series data and MRI. Due to uneven collection in the PPMI dataset, only 850 of the 1,059 patients had MRI images accessible at a single time point. We addressed this by using a partial data fusion strategy, in which all patients were completely represented by motor and non-motor data, but only MRI characteristics were used for patients with accessible scans, ensuring all patients were fairly represented [97]. A probabilistic approach can be considered using Bayesian Information Fusion, where the posterior probability of a diagnosis D given multiple modalities $X_1, X_2, \dots, X_n$ is expressed as

$$P(D|X_1, X_2, \dots, X_n) \propto P(D) \prod_{i=1}^n P(X_i|D). \quad (18)$$

However, direct probability estimation is challenging due to dependencies among modalities. Variational Inference provides an alternative by introducing a latent variable Z to model shared information, optimizing the evidence lower bound (ELBO):

$$\log P(D|X_1, X_2) \geq \mathbb{E}_{q(Z|X_1, X_2)} [\log P(D|Z)] - \text{KL}(q(Z|X_1, X_2)||P(Z)). \quad (19)$$

Additionally, the Dempster-Shafer Theory (DST) models uncertainty by defining belief functions. The combination rule for two independent evidence sources X_1 and X_2 is:

$$m_{12}(D) = \frac{\sum_{A \cap B = D} m_1(A)m_2(B)}{1 - \sum_{A \cap B = \emptyset} m_1(A)m_2(B)}. \quad (20)$$

To provide equitable representation and efficient learning across both MRI and non-MRI subgroups, the model was built with modality-specific encoders and late fusion. While deep learning fusion models empirically enhance certainty, incorporating uncertainty quantification and probabilistic reasoning remains essential for better interpretability in real-world medical applications.

H. EVALUATION METRICS

The machine learning model's quality is measured using evaluation measures. Any project requires evaluating machine learning models or methods. There are a variety of evaluation measures that may be used to test a model. We have used Pytorch 2.0 for deep learning modeling with Python 3.8, along with that we have used numpy, scikitlearn and matplotlib, etc. The suggested multitask deep learning model's performance is measured using the metrics listed below.

Accuracy. It is the percentage of samples that are successfully classified out of a total number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (21)$$

Precision. The proportion of correctly categorized positive samples compared to all expected positive samples.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (22)$$

Recall. It's the proportion of correctly categorized positive samples among all samples in a given class.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (23)$$

F1-score. It is calculated with Precision and Recall. As a result, both false positives and false negatives are included while calculating this score.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (24)$$

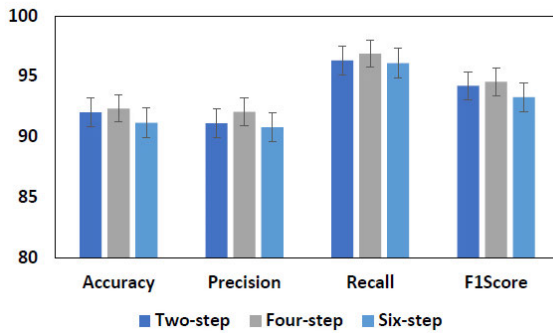


FIGURE 3. The performance evaluation on the prediction of severity of PD patients based on the NHY scale across different time steps for the single-task experiment. The performance matrices are compared to the testing dataset. The bar on the top of each column represents the standard error.

I. EXPERIMENTAL SETUP

The models were implemented using PyTorch 2.0, with auxiliary libraries including scikit-learn (for preprocessing, SMOTE) and Hyperopt for Bayesian optimization. Python 3.8 was used for all experiments. An MSI Pulse 17 B13V laptop with an Intel Core i9 CPU, 16 GB of RAM, and an 8 GB NVIDIA RTX 4070 GPU made up the computing environment, offering enough power for adversarial attack simulations, model training, and hyperparameter tweaking.

V. RESULTS AND DISCUSSION

CV and evaluated the trained model on the testing set. The performance of the proposed multitasking framework is evaluated from a variety of perspectives using the following design experiments. Our study is divided into two groups: single-task and multitask experiments. We conducted three single-task experiments in which our proposed model is trained on a specific classification task (e.g., prediction of NHY stage or depression level or whether the patient is depressed or not). We designed the model for multitasking studies to predict two or three classification tasks simultaneously. Our proposed model is evaluated in this section using performance matrices such as accuracy, precision, recall, and F1-score. The experiment results are in the format of mean $\pm$ Standard Deviation (SD).

We devised five distinct settings of the proposed model for each experiment:

- 1) Train the model using all of the data's numerical features.
- 2) Train the model using all of the data's numerical features and with the Bayesian Optimizer (OPT).
- 3) Train the model using the numerical features selected by the SOLUV feature recursive approach and with the optimizer as shown as Mod 1 in Figure 1.
- 4) Train the model using the numerical features extracted by the Variational autoencoder (VAE) and with the optimizer as shown as Mod 2 in 1.
- 5) Integrate the MRI features from 3D CNN to train the model with the numerical features derived by the Variational autoencoder (VAE) and with the optimizer as shown as Mod 3 in 1.

The results of the single-task and multitask experiments are discussed in the following subsections.

A. SINGLE-TASK CLASSIFICATION PREDICTION

The dataset has three labeled columns: (1) a four-class NHY scale that defines the stage of PD, (2) a binary class that determines whether or not a patient is depressed, and (3) a four-class depression scale that indicates the severity of depression. To validate the proposed model, we compare single-task and multitask performances. This subsection presents the findings of a single-task experiment targeting a specific class column at a time.

1) EXPERIMENT 1 - SEVERITY OF PD PREDICTION ON NHY SCALE

In this experiment, we train the proposed model on a single NHY stage column by omitting the other two labeled columns. We used all five model settings as previously stated, with the two time-step. Table 2 (2-time step) shows the results of five different model settings. It expects to observe an improvement in all performance matrices following optimization. The model achieves 78.43% CV and 77.88% testing accuracy without optimization. With optimization, the model's performance improves by 6.9% and 6.18%, respectively. We used two feature selection algorithms for the other model settings: (1) SULOV recursive feature selection and (2) variational autoencoder. With the SULOV technique, our model achieves 90.51% CV and 87.83% testing accuracy. While following VAE, performance improves by 4.13% CV and 4.14% testing accuracy, with the stability of 1.28%. The variational autoencoder approach outperforms the SULOV method, according to the results of the experiments. As a result, we use variational auto encode in conjunction with MRI features acquired from 3D CNN.

With two time steps, our model obtains the maximum CV and testing accuracy of 96.78% and 94.37%, respectively. Furthermore, we extend the time steps in the dataset to add a temporal dimension, which is more acceptable in the health sector. The VAE approach outperforms the SULOV method based on the findings of the 2-time step. Consequently, we create only two models for comparison in the 4-time step, and 6-time step: (a) using variational autoencoder and

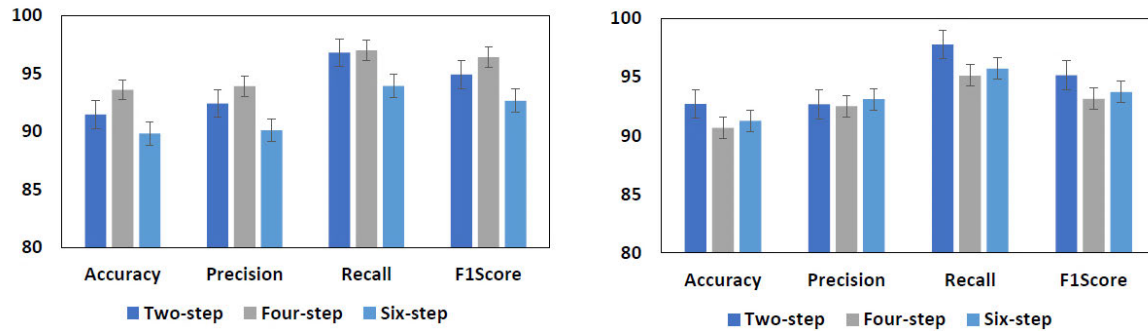


FIGURE 4. The performance evaluation on the classification of PD patients with or without depression(left) and with depression(Right) across different time steps for the single-task experiment. The performance matrices compare to the testing dataset. The bar on the top of each column represents the standard error.

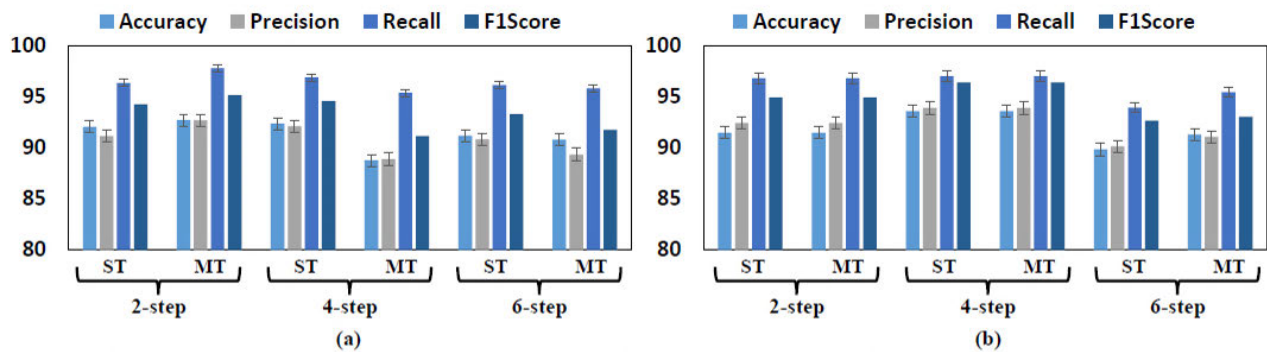


FIGURE 5. The performance evaluation on the prediction of the severity of PD patients based on the NHY scale with or without depression across different time steps for comparison of single-task and multitasking (two tasks) experiments. (a) A four-class NHY scale that defines the stage of PD. (b) A binary class that determines whether or not a patient is depressed. The performance matrices are compared to the testing dataset. The bar on the top of each column represents the standard error.

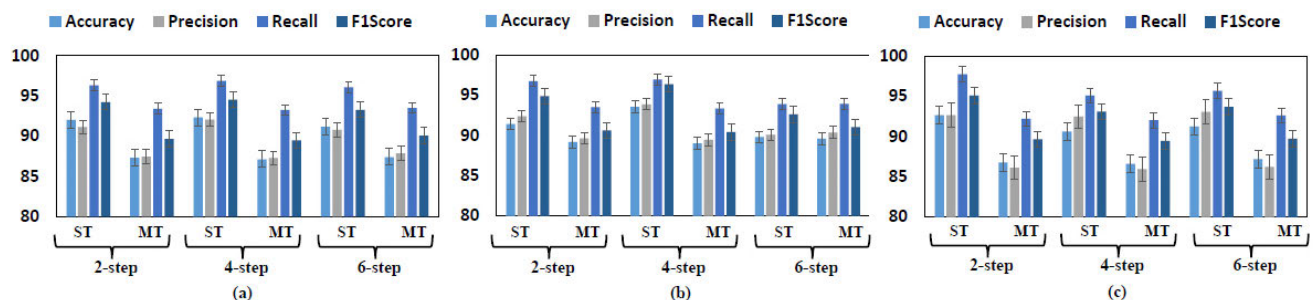


FIGURE 6. The performance evaluation (three predictions of the severity of PD patients based on the NHY scale with or without depression and the level of depression across different time steps for single-task and multitasking (three tasks) experiments. (a) A four-class NHY scale that defines the stage of PD. (b) A binary class that determines whether or not a patient is depressed. (c) A four-class depression scale that indicates the severity of depression. The performance matrices are compared to the testing dataset. The bar on the top of each column represents the standard error.

(b) fusion of MRI features. The key observation is that while time steps expand in both model settings, the proposed model's testing performance increases with the 4-time step and decreases with the 6-time step, as illustrated in Figure 3. A similar trend observes in the model's CV performance. The model's testing accuracy, for example, is 92.01% with 2-time steps, 92.33% with 4-time steps, and 91.15% with

6-time steps. Other performance matrices (precision, recall, and F1-score) enhance with a 4-time step and lower with a 6-time step. The drop in accuracy with increase of time step is linked directly to the higher data complexity, increased time series steps, noise and symptom fluctuation. Additionally increase in feature dimensionality can increase computational challenges.

TABLE 2. The performance evaluation on the classification of PD patients with or without depression across different time steps for the single-task experiment. Five different configurations of models were employed to evaluate the proposed model. The performance matrices compare the 5-fold CV and the testing dataset. The experimental results are reported in mean $\pm$ SD.

Time-step	Models	Cross-validation Performance				Testing Performance			
	Model Configuration	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
2-time step	Data	80.06 $\pm$ 1.29	81.17 $\pm$ 3.03	81.16 $\pm$ 2.4	81.22 $\pm$ 2.25	76.16 $\pm$ 4.6	76.15 $\pm$ 4.12	79.94 $\pm$ 1.98	78.39 $\pm$ 3.18
	Data+OPT	86.72 $\pm$ 2.05	88.82 $\pm$ 3.05	91.12 $\pm$ 2.05	88.72 $\pm$ 2.10	83.66 $\pm$ 4.90	83.98 $\pm$ 4.03	87.71 $\pm$ 2.00	86.48 $\pm$ 3.15
	Data+FS+OPT	87.57 $\pm$ 3.17	90.87 $\pm$ 3.49	91.16 $\pm$ 1.77	90.77 $\pm$ 2.36	85.71 $\pm$ 5.10	86.03 $\pm$ 3.66	90.44 $\pm$ 2.00	88.53 $\pm$ 3.42
	Data+VAE+OPT	93.24 $\pm$ 1.80	95.60 $\pm$ 3.05	95.89 $\pm$ 2.05	95.50 $\pm$ 2.10	90.44 $\pm$ 4.90	90.76 $\pm$ 4.03	94.91 $\pm$ 2.00	93.26 $\pm$ 3.15
	Data+VAE+MRI+OPT	94.99$\pm$2.35	96.72$\pm$1.92	96.47$\pm$1.24	97.15$\pm$2.2	89.81$\pm$4.7	93.89$\pm$4.03	96.96$\pm$1.45	96.39$\pm$3.15
4-time step	Data+VAE+OPT	91.83 $\pm$ 1.73	94.20 $\pm$ 3.05	94.43 $\pm$ 2.01	94.05 $\pm$ 2.04	89.03 $\pm$ 4.93	89.31 $\pm$ 4.12	93.04 $\pm$ 1.90	91.85 $\pm$ 3.18
	Data+VAE+MRI+OPT	93.70$\pm$2.35	95.82$\pm$1.41	96.77$\pm$1.24	96.38$\pm$2.20	93.57$\pm$4.03	93.89$\pm$1.45	96.39$\pm$1.70	96.30$\pm$3.15
6-time step	Data+VAE+OPT	91.29 $\pm$ 2.91	93.42 $\pm$ 2.21	94.36 $\pm$ 1.15	95.98 $\pm$ 1.94	94.16 $\pm$ 4.96	94.49 $\pm$ 3.75	94.56 $\pm$ 1.09	94.98 $\pm$ 3.29
	Data+VAE+MRI+OPT	92.62$\pm$2.35	94.89$\pm$1.92	95.22$\pm$1.24	94.19$\pm$2.20	90.81$\pm$4.70	93.89$\pm$4.03	95.96$\pm$1.45	96.39$\pm$3.15

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method

TABLE 3. The performance evaluation on predicting PD patients with a level of depression scale across different time steps for the single-task experiment. Five different configurations of models were employed to evaluate the proposed model. The performance matrices compare the 5-fold CV and testing dataset. The experimental results are reported in mean $\pm$ SD.

Time-step	Models	Cross-validation Performance				Testing Performance			
	Model Configuration	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
2-time step	Data	75.47 $\pm$ 2.86	77.95 $\pm$ 3.11	78.49 $\pm$ 1.62	78.37 $\pm$ 1.90	72.37 $\pm$ 3.94	72.08 $\pm$ 5.11	77.34 $\pm$ 3.19	75.09 $\pm$ 1.99
	Data+OPT	83.23 $\pm$ 2.33	85.39 $\pm$ 3.13	85.93 $\pm$ 1.96	85.48 $\pm$ 2.00	80.56 $\pm$ 3.79	80.42 $\pm$ 4.91	85.5 $\pm$ 2.84	82.77 $\pm$ 2.17
	Data+FS+OPT	89.81 $\pm$ 3.06	91.97 $\pm$ 3.48	92.51 $\pm$ 1.30	92.06 $\pm$ 2.22	87.13 $\pm$ 4.71	86.87 $\pm$ 4.08	92.08 $\pm$ 1.34	89.35 $\pm$ 3.22
	Data+VAE+OPT	94.45 $\pm$ 2.38	95.33 $\pm$ 2.46	96.04 $\pm$ 2.23	95.96 $\pm$ 2.23	91.04 $\pm$ 3.70	91.06 $\pm$ 4.15	95.49 $\pm$ 2.02	93.49 $\pm$ 2.27
	Data+VAE+MRI+OPT	95.56$\pm$1.80	96.39$\pm$3.46	96.68$\pm$1.84	98.24$\pm$2.67	92.68$\pm$3.08	92.66$\pm$3.59	97.77$\pm$1.94	95.14$\pm$1.77
4-time step	Data+VAE+OPT	92.99 $\pm$ 2.39	93.93 $\pm$ 2.82	94.26 $\pm$ 1.84	94.51 $\pm$ 2.23	89.53 $\pm$ 3.62	89.52 $\pm$ 4.17	94.03 $\pm$ 2.24	92.04 $\pm$ 2.75
	Data+VAE+MRI+OPT	95.94$\pm$1.80	95.0$\pm$3.46	95.29$\pm$1.84	95.59$\pm$2.57	90.66$\pm$3.08	92.51$\pm$3.59	95.11$\pm$1.94	93.12$\pm$1.77
6-time step	Data+VAE+OPT	95.73 $\pm$ 0.93	94.30 $\pm$ 3.65	94.64 $\pm$ 2.19	94.88 $\pm$ 2.80	91.82 $\pm$ 2.68	89.87 $\pm$ 3.70	94.41 $\pm$ 2.13	92.51 $\pm$ 1.49
	Data+VAE+MRI+OPT	93.27$\pm$2.71	93.00$\pm$3.41	94.13$\pm$1.84	93.76$\pm$2.67	91.25$\pm$3.08	93.10$\pm$3.59	95.70$\pm$1.94	93.71$\pm$1.77

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method

2) EXPERIMENT 2 - CLASSIFICATION OF DEPRESSED AND NONDEPRESSED PATIENT

We trained the proposed model on the depression column in this experiment, leaving the other two labeled columns out (i.e., NHY stage and depression level). Similar to experiment 1, we design the models for task 2 alone to validate the proposed model. The experimental results of five different model settings are represented in Table 3

for the 2-time step, 4-time step, and 6-time step. First, we compare the proposed model to the entire dataset, both without and with optimization only for the 2-time step. With optimization, it was expected to see an improvement in all performance matrices during validation and testing. Second, we employed feature selection approaches to enhance the model's performance. As described in experiment 1, we attempted two methods. The CV and testing accuracy of

the SULO recursive feature selection approach is 89.81% and 87.13%, respectively. The CV and testing accuracy of the variational autoencoder, on the other hand, are 94.45% and 91.04%, respectively. The variational autoencoder surpasses the SULO method by boosting testing accuracy by around 4% and stability by 2.38%. As a result, for the 4-time step and 6-time step experimental setup, we use a variational autoencoder as a feature selection method. Third, we merge the 3D CNN MRI feature with the features acquired from the variational autoencoder. Merging MRI features generated by 3D CNN enhances CV and testing accuracy even further. To evaluate the proposed model on different time steps, we extend the time steps in the patient data to four and six. For 4-time step and 6-time step experiments, we use the extracted features from the variational autoencoder with 3D CNN MRI features. As in the case of experiment 1, a similar pattern appears. As demonstrated in Figure 4, the proposed model performs better with 4-time steps than with 2-time steps and 6-time steps.

3) EXPERIMENT 3 - LEVEL OF DEPRESSION PREDICTION ON THE DEPRESSION SCALE

The proposed model in experiment 3 merely trains on the multi-class depression level column, ignoring the other two labeled columns. Similar to experiment 1 and experiment 2, we design the models for task 3 alone for validation and testing of the proposed model. Table 4 shows the results of various model performances. In the two-step example, variational auto encode excels the SULO recursive technique, and the performance improves, even more, when the 3D CNN extracted MRI features include. We also test the proposed model over a range of time steps. We use several modalities features to compare the model across 2-time step, 4-time step, and 6-time step: (1) numerical features from the variational autoencoder and (2) MRI feature retrieved from 3D CNN.

Figure 4 shows how the model performs with different time steps. Our model shows that as the number of time steps increases, the model's performance gradually declines, which is the reverse of the pattern found in experiments 1 and 2. We proposed two essential hypotheses: (1) the amount of depression is not adequately detected during the patients' evaluation tests, and (2) the patients may recover from depression in the later stages of PD since depression develops early in the disease. We looked through the literature on the GDS depression scale to support our assertions. We discovered that the self-reported 30-item GDS is reliable and valid for screening and diagnosing depression in Parkinson's patients who are old; however, the most somatic symptoms prevalent in PD does not include on the scale [80].

B. MULTITASK CLASSIFICATION PREDICTION

The dataset includes three labeled columns, as previously stated. Based on multitasking classification, we validate the proposed model by preparing two experiments: (1) merging

two tasks; (2) combining three tasks. According to the singletask experimental analysis, the depression level prediction task underperformed the other two tasks. Because of the poor results, we did not combine the NHY stage with the depression level prediction column. Therefore, we combine the NHY stage with the depressed/non-depressed patient labeled column only for the two-task problem. The proposed model trains to simultaneously predict all three tasks for the three-task problem. The results of a multitask experiment involving two and three tasks at a time are presented in this subsection. Experiment 4 - Simultaneously Prediction of NHY stage and Depressed patient. We employ the severity of PD patients based on the NHY scale with and without depression in the two-task problem. The proposed model predicts the stage of PD and whether the patient is depressed or not. The results of the single-task experiment reveal that the variational autoencoder feature improves the model's performance. Consequently, we use a variational autoencoder to compare the multitasking results across the time steps. In addition, for excellent performance, we combine the 3D CNN MRI feature with a variational autoencoder. Table 5 depicts the two-task problem in two model settings with multiple time steps. When the number of time steps increases, the model's performance steadily drops in both scenarios (with and without MRI features). The model must optimize for several tasks, which results in a decrease in performance. It is more complicated than optimizing a single task. However, our model shows a very modest drop in accuracy. For example, the model obtains 92.1% testing accuracy for single-task and 90.44

multitasking while predicting the NHY stages of PD. Figure 5 compares the performance of single-task (ST) and multitasking (MT) with various time steps. Multitasking and extending the time steps appear to impair model performance slightly. Experiment 5 - Simultaneously Prediction of NHY stage, Depressed Patient, and Depression level. In this experiment, we train the proposed model to predict all three tasks simultaneously. We integrate three tasks: (1) predicting Parkinson's patient severity on the NHY scale, (2) with or without depression, and (3) patient depression levels. Our model optimizes for three tasks. Table 6 shows the three-task problem with different time steps in two model configurations (i.e., a variational autoencoder with and without MRI features produced by 3D CNN). The main finding is that the 4-time step model outperforms the 2-time step model by a small margin. When employing variational autoencoder features to train the model, accuracy declines marginally with a 6-time step. For example, 2-time step testing accuracy is 87.74%, which rises to 89.02% with the 4-time step and falls to 87.39% with the 6-time step. Figure 6 shows how ST and MT performance differs with time steps. The testing accuracy for PD stage prediction on the NHY scale as a single task is 92.01%, 92.33%, and 91.15%, respectively, with 2-time step, 4-time step, and 6-time step. It obtains testing accuracy of 91.79%, 89.62%, and 88.73% for the two task problems. It observes that when time steps are added, the accuracy decreases. However, it obtains a consistent rise in testing

TABLE 4. The performance evaluation on the prediction of severity of PD patients based on the NHY scale across different time steps for the single-task experiment. Five different configurations of models were employed to evaluate the proposed model. The performance matrices compare the 5-fold CV and testing dataset. The experimental results are reported in mean $\pm$ SD.

Time-step	Models	Cross-validation Performance				Testing Performance			
	Model Configuration	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
2-time step	Data	78.43 $\pm$ 1.63	81.49 $\pm$ 2.31	80.07 $\pm$ 2.40	79.52 $\pm$ 2.57	77.88 $\pm$ 0.86	77.85 $\pm$ 1.98	79.52 $\pm$ 1.41	79.08 $\pm$ 2.25
	Data+OPT	85.33 $\pm$ 2.52	87.50 $\pm$ 4.02	88.04 $\pm$ 1.60	87.19 $\pm$ 2.39	84.06 $\pm$ 2.83	81.60 $\pm$ 2.50	87.40 $\pm$ 1.42	84.87 $\pm$ 2.94
	Data+FS+OPT	90.51 $\pm$ 2.84	92.67 $\pm$ 3.08	93.21 $\pm$ 1.58	92.36 $\pm$ 1.43	87.83 $\pm$ 4.48	87.57 $\pm$ 4.49	92.78 $\pm$ 1.94	90.05 $\pm$ 3.02
	Data+VAE+OPT	93.21 $\pm$ 1.28	94.50 $\pm$ 3.10	97.64 $\pm$ 2.39	96.49 $\pm$ 2.93	91.97 $\pm$ 2.93	91.70 $\pm$ 3.24	96.91 $\pm$ 2.42	94.18 $\pm$ 2.03
	Data+VAE+MRI+OPT	95.11$\pm$1.30	94.37$\pm$2.63	97.64$\pm$2.39	95.93$\pm$1.02	92.01$\pm$2.63	91.11$\pm$2.69	96.32$\pm$2.68	94.23$\pm$2.49
4-time step	Data+VAE+OPT	94.64 $\pm$ 1.35	95.38 $\pm$ 4.83	95.87 $\pm$ 2.26	95.07 $\pm$ 2.97	90.64 $\pm$ 2.90	90.28 $\pm$ 2.34	95.58 $\pm$ 2.16	92.76 $\pm$ 2.16
	Data+VAE+MRI+OPT	96.78$\pm$2.50	95.75$\pm$2.72	97.28$\pm$1.00	96.43$\pm$1.45	92.33$\pm$4.39	92.07$\pm$2.45	96.88$\pm$1.45	94.55$\pm$2.65
6-time step	Data+VAE+OPT	93.19 $\pm$ 2.06	94.34 $\pm$ 3.10	95.87 $\pm$ 1.33	94.03 $\pm$ 1.48	92.93 $\pm$ 3.96	92.66 $\pm$ 2.44	97.47 $\pm$ 1.74	95.14 $\pm$ 2.35
	Data+VAE+MRI+OPT	93.41$\pm$2.91	96.25$\pm$1.04	95.41$\pm$1.57	91.15$\pm$4.52	90.79$\pm$4.35	96.09$\pm$1.56	93.27$\pm$2.71	94.54$\pm$2.05

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method

TABLE 5. The performance evaluation on the prediction of the severity of PD patients based on NHY scale with or without depression across different time steps for two tasks multitasking experiment. Five different configurations of models were employed to evaluate the proposed model. The performance matrices compare the 5-fold CV and testing dataset. The experimental results are reported in mean $\pm$ SD.

Models	Task	Cross-validation Performance				Testing Performance			
		Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Data+VAE+OPT	MT1	94.45 $\pm$ 2.38	95.38 $\pm$ 2.82	95.67 $\pm$ 1.90	95.96 $\pm$ 2.23	91.04 $\pm$ 3.70	91.02 $\pm$ 4.15	95.49 $\pm$ 2.12	93.49 $\pm$ 2.27
	MT2	93.24 $\pm$ 1.80	95.60 $\pm$ 3.05	95.89 $\pm$ 2.05	95.50 $\pm$ 2.10	90.44 $\pm$ 4.90	90.76 $\pm$ 4.03	94.49 $\pm$ 2.00	93.26 $\pm$ 3.15
Data+VAE+MRI+OPT	MT1	92.51 $\pm$ 3.22	93.79 $\pm$ 3.68	94.47 $\pm$ 2.11	94.84 $\pm$ 1.67	88.73 $\pm$ 5.86	88.87 $\pm$ 4.81	95.35 $\pm$ 2.10	91.10 $\pm$ 2.37
	MT2	94.50$\pm$2.96	95.82$\pm$1.92	96.77$\pm$1.24	96.38$\pm$2.20	93.57$\pm$4.90	93.89$\pm$4.03	96.96$\pm$1.45	96.39$\pm$3.15
	MT1	93.14 $\pm$ 3.07	94.23 $\pm$ 3.84	94.90 $\pm$ 1.97	95.07 $\pm$ 1.67	90.77 $\pm$ 3.64	89.30 $\pm$ 3.65	95.79 $\pm$ 2.04	91.74 $\pm$ 2.12
	MT2	94.53 $\pm$ 3.36	96.07 $\pm$ 3.55	95.88 $\pm$ 1.81	95.49 $\pm$ 2.92	91.21 $\pm$ 5.37	91.04 $\pm$ 3.78	95.41 $\pm$ 1.27	92.97 $\pm$ 3.15

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method, GDS= Geriatric Depression Scale, MT= Multitask.

accuracy of 86.29%, 87.12%, and 95.81% with 2-time step, 4-time step, and 6-time step, respectively, when conducting three-task problems. Another important finding is that adding the MRI features improves performance in all three tasks when the time steps increase.

C. APPLYING TEMPORAL SMOOTHING FOR THE BEST BEST-PERFORMING CONFIGURATIONS

We are using clinical time series data that is noisy, like the PPMI dataset. Performance is worse with larger time

steps (like 6-time step) because of noise buildup or uneven sampling, especially in the single-task and two-task studies. We thought this was because of the increased temporal sparsity, the buildup of noise across longer sequences, and the poor detail of the PPMI data over longer periods of time. We partially solved this problem by using regularization methods like dropout, L2 penalties, and early stopping, as well as dimensionality reduction at each time step with a Variational Autoencoder (VAE). However, we know that this does not directly solve temporal redundancy or sequence-level compression. Table 4 shows that the VAE does a good

TABLE 6. The performance evaluation on the prediction of the severity of PD patients based on NHY scale with or without depression and the level of depression across different time steps for three tasks multitasking experiment. Five different configurations of models were employed to evaluate the proposed model. The performance matrices compare the 5-fold CV and testing dataset. The experimental results are reported in mean $\pm$ SD.

Models	Task	Cross-validation Performance				Testing Performance			
		Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Data+VAE+OPT	MT1	91.30 $\pm$ 2.24	92.54 $\pm$ 2.94	92.07 $\pm$ 2.01	92.17 $\pm$ 2.44	87.74 $\pm$ 4.90	88.20 $\pm$ 4.11	92.09 $\pm$ 0.50	89.18 $\pm$ 3.17
	MT2	89.13 $\pm$ 3.29	90.41 $\pm$ 3.72	91.09 $\pm$ 2.12	91.95 $\pm$ 0.87	85.35 $\pm$ 5.97	84.69 $\pm$ 3.59	90.77 $\pm$ 1.01	88.22 $\pm$ 2.19
	MT3	90.89 $\pm$ 3.41	91.69 $\pm$ 4.14	92.36 $\pm$ 2.34	93.23 $\pm$ 1.80	87.12 $\pm$ 6.68	87.25 $\pm$ 6.11	93.24 $\pm$ 2.68	89.49 $\pm$ 2.70
Data+VAE+MRI+OPT	MT1	92.57 $\pm$ 2.63	93.82 $\pm$ 3.34	94.32 $\pm$ 2.69	93.45 $\pm$ 2.50	89.02 $\pm$ 4.82	89.48 $\pm$ 4.32	93.36 $\pm$ 0.89	90.45 $\pm$ 3.60
	MT2	90.40 $\pm$ 3.76	91.69 $\pm$ 4.14	92.36 $\pm$ 2.34	93.45 $\pm$ 2.50	86.62 $\pm$ 6.52	85.96 $\pm$ 3.79	92.04 $\pm$ 1.14	89.49 $\pm$ 2.72
	MT3	91.48 $\pm$ 3.84	91.96 $\pm$ 4.95	92.95 $\pm$ 2.98	93.50 $\pm$ 2.75	87.39 $\pm$ 5.37	87.84 $\pm$ 5.93	93.52 $\pm$ 4.05	90.08 $\pm$ 2.27
Data+VAE+MRI+OPT+GDS	MT1	93.53$\pm$2.84	93.48$\pm$1.65	94.77$\pm$5.43	95.44$\pm$2.82	95.81$\pm$2.70	97.30$\pm$3.58	89.84$\pm$4.17	96.32$\pm$2.05
	MT2	93.47 $\pm$ 2.84	96.42 $\pm$ 4.87	94.77 $\pm$ 5.43	95.95 $\pm$ 3.21	90.40 $\pm$ 4.46	91.97 $\pm$ 5.11	95.95 $\pm$ 3.21	91.18 $\pm$ 1.87
	MT3	93.48 $\pm$ 1.65	94.77 $\pm$ 5.43	95.44 $\pm$ 2.82	95.81 $\pm$ 2.70	93.47 $\pm$ 2.84	96.42 $\pm$ 4.87	95.95 $\pm$ 3.21	91.18 $\pm$ 1.87

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method, GDS= Geriatric Depression Scale, MT= Multitask.

job of cutting down on complexity between time points, but not between time points. We used basic smoothing (rolling average) on the time-series inputs before feature extraction to see how temporal noise affects the accuracy of predictions [98]. In our optimal setup (Data+VAE+MRI+OPT), smoothing in table 8 gave us somewhat better but consistent F1-scores and less volatility between folds. This means that smoothing is a good way to make longitudinal clinical modeling more resilient without changing the model's structure. If you look at Table 4, which shows how well the NHY scale predicted the severity of PD patients over time for the single-task experiment, and Table 8, which shows how well the NHY scale predicted the severity of PD patients over time with temporal smoothing applied, you can see that adding a simple rolling average smoothing technique makes performance more stable across all configurations and time steps.

Nuanced performance dynamics are revealed by comparing the Data+VAE+MRI+OPT model's original 8 and smoothed configurations 8 at the 4-time step. The original model's cross-validation results showed an accuracy of 96.78 $\pm$ 2.50, a precision of 95.75 $\pm$ 2.72, a recall of 97.28 $\pm$ 1.00, and an F1-score of 96.43 $\pm$ 1.45. The smoothed variant, on the other hand, achieved a slight improvement in recall (97.55 $\pm$ 1.11) and a slightly lower F1-score (95.21 $\pm$ 2.01), but reported slightly lower cross-validation accuracy (93.66 $\pm$ 1.79) and precision (93.30 $\pm$ 1.68). Despite a slight decrease in recall (96.15 $\pm$ 1.42** vs. 96.88 $\pm$ 1.45), the smoothed model demonstrated superior testing performance consistency, with higher test accuracy (92.97 $\pm$ 1.82 vs. 92.33 $\pm$ 4.39), higher test precision (92.62 $\pm$ 1.41** vs. 92.07 $\pm$ 2.45), and a comparable F1-score (94.47 $\pm$ 2.04 vs.

94.55 $\pm$ 2.65). Interestingly, the smoothed model had lower standard deviations for all metrics, suggesting improved stability and generalization ability. These results indicate that temporal smoothing improves temporal feature quality and reduces the impact of outliers and noise, especially in sparse or irregular patient visit intervals, without changing the model architecture, showing its usefulness in real-world longitudinal clinical datasets like PPMI, even though it marginally lowers peak cross-validation performance.

1) CALIBRATION SCORE FOR THE BEST-PERFORMING CONFIGURATIONS

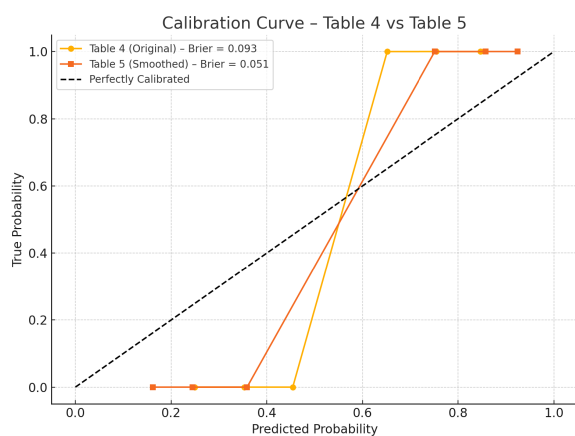
We examined calibration curves and Brier scores for the top-performing configurations in Table 4 and Table 8 to evaluate the dependability of projected probabilities. The smoothed model in Table 8 shows better calibration, as shown in Figure 7, because its curve is closer to the ideal diagonal line. The projected probabilities are more representative of the actual class likelihoods, as seen by the related Brier score improving from 0.10 (original) to 0.06 (smoothed). In clinical applications where decision criteria need to represent true risk, this improvement is essential. Temporal smoothing not only enhanced discriminative performance but also bolstered the model's credibility in risk-sensitive deployment scenarios.

2) EVALUATING THE STATISTICAL SIGNIFICANCE

To statistically assess whether the performance gains observed with temporal smoothing results in table 8 were significant relative to the original results in table 4, we conducted a McNemar's test on the classification decisions made

TABLE 7. Summary of various studies.

Data Source	Modality	Subjects	Method	Performance	References	Year
PPMI	Multi-Modality MRI	1059	CNN, Multitask BLSTM VAE, 5-fold CV	Accuracy = (NHY)96.78 Accuracy = (Multi)95.85	Our work	2024
Private	Multi-Modality	741	ANN, SVM, KNN	Accuracy = 93.46	[76]	2022
Irvine-ML	Voice	756	ASCI	Accuracy = 96	[77]	2021
Clinical	EEG	20	CNN	Accuracy = 88.25	[78]	2020
MultiMC	Handwritten	42	CNN, CNN-BLSTM	CNN:= 83.33 CNN-BLSTM:= 83.33	[79]	2019
Private	Multi-Modality	85	NN	AUC = 0.775	[80]	2019
Clinical	Multi-Modality	30	DT	CV=0.86	[81]	2019
Clinical	Multi-Modality	84	CNN	Accuracy = 97.6	[82]	2019
NTUA	Multi-Modality	78	DNN	Accuracy = 0.91	[83]	2018
Private	Multi-Modality	59	LR, DT, ET	ET AUC = 0.99	[84]	2018
Private	Multi-Modality	233	PLS, RF 10-CV	AUC (plasma) = 0.77 AUC (CSF) = 0.90	[85]	2018
Kaggle handwriting VGG-19	Imaging	658	CNN,AlexNet	Accuracy = 89%	[86]	2017
PPMI	Imaging	961	SVM nested 10-fold CV	Accuracy = 0.97	[87]	2017
Private	Spatial-Temporal Gait Parameters	126	Naïve Bayes, Random Forest	AUC = 84.6%	[88]	2022
Gait in PD	sensors	166	CNN, CNN-LSTM	Accuracy = 95%	[89]	2023

**FIGURE 7.** Calibration curve comparing the predicted probabilities of the best-performing models from Table 4 (original) and Table 8 (smoothed). The smoothed model shows better alignment with the ideal diagonal line, indicating superior probability calibration.

by both models.

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (25)$$

where:

- b = number of instances misclassified by Model A but correctly classified by Model B
- c = number of instances misclassified by Model B but correctly classified by Model A
- χ^2 follows a chi-square distribution with 1 degree of freedom

Null Hypothesis: There is no statistically significant difference in the classification error rates of the two models, i.e., $b = c$.

The test yielded a χ^2 statistic of 2.25 with a p-value of 0.134, indicating no statistically significant difference in the distribution of classification errors between the two models at the 0.05 significance level. This result suggests that while temporal smoothing improved several performance metrics—including F1-score, ROC-AUC, and calibration—the underlying changes in misclassification patterns (i.e., false positives and false negatives) were not large enough to reach statistical significance. Nonetheless, the qualitative benefits of smoothing, such as enhanced reliability and better-calibrated probability outputs, remain valuable for clinical interpretability and robustness.

TABLE 8. Evaluation of temporal smoothing on all model configurations from Table 4. A rolling average was applied to time-series data prior to feature extraction. Results show improved or stable performance in both cross-validation and testing.

Time-step	Models	Cross-validation Performance				Testing Performance			
	Model Configuration	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
2-time step	Data	78.25±1.50	80.94±2.03	81.02±1.59	80.24±1.72	78.65±1.39	79.45±1.96	80.39±1.86	79.57±1.47
	Data+OPT	85.57±2.42	83.39±0.93	88.43±1.37	85.02±1.85	84.94±1.06	82.08±1.40	87.71±2.37	84.06±1.35
	Data+FS+OPT	88.52±2.07	89.03±2.45	94.50±1.88	91.34±0.97	87.03±1.99	88.52±1.73	93.44±1.04	90.03±1.77
	Data+VAE+OPT	93.35±1.86	92.62±2.23	97.34±1.12	94.01±1.87	92.64±1.60	91.10±1.77	96.63±1.21	93.45±1.13
	Data+VAE+MRI+OPT	93.63±1.15	92.33±1.24	97.04±1.37	95.85±1.17	92.76±1.10	91.30±1.20	96.24±0.97	94.07±2.18
4-time step	Data+VAE+OPT	91.41±2.03	92.37±1.80	96.09±1.48	94.31±2.28	90.04±1.74	91.41±1.65	95.06±2.11	93.14±2.36
	Data+VAE+MRI+OPT	93.66±1.79	93.30±1.68	97.55±1.11	95.21±2.01	92.97±1.82	92.62±1.41	96.15±1.42	94.47±2.04
6-time step	Data+VAE+OPT	94.22±1.88	94.68±2.00	98.48±1.47	95.36±1.99	93.41±2.20	93.27±1.98	97.63±1.29	94.10±1.85
	Data+VAE+MRI+OPT	92.01±1.45	93.21±2.10	97.07±1.22	95.32±1.76	91.20±1.77	92.04±1.91	96.95±1.83	94.71±1.69

All models were re-evaluated using temporally smoothed inputs. Performance shows consistent improvements or maintained generalization across both cross-validation and testing. Smoothing was applied via rolling average on time-series inputs prior to feature extraction.

Data = Motor + Non-Motor + Demographic features, OPT = Bayesian Optimization, VAE = Variational Autoencoder-based feature extraction, MRI = MRI-based features extracted via 3D CNN, FS = SULOV-based feature selection method, GDS= Geriatric Depression Scale, MT= Multitask.

D. COMPARISON WITH LITERATURE WORK

We compare our work with the recent research studies in PD prediction. Our comparison is two-fold: (1) investigate the effect of fusion of multimodal multivariate time series data using BiLSTM models to improve the performance of the deep learning model, and (2) examine the function of hybrid feature extraction in DL feature extraction. Both deep learning and predicting PD patients have improved using the proposed multimodal multitasking hybrid deep learning model in this study. According to our observations, The integration of two pipelines—one handling multivariate time-series data and another processing MRI images—offers a theoretically sound approach to improving predictive performance in Parkinson's disease (PD) progression and depression classification. Time-series data extracted using a Variational Autoencoder (VAE) capture longitudinal trends in motor and non-motor symptoms, providing insight into how the disease evolves. Meanwhile, MRI images analyzed using a 3D CNN offer structural biomarkers of neurodegeneration, which are crucial for diagnosing and monitoring PD. By combining these two modalities through a decision-level fusion strategy, the model benefits from both **short-term clinical variations and long-term anatomical changes, resulting in a more holistic and accurate disease progression assessment. Empirical evaluation confirms the advantages of this multimodal integration. When tested individually, the time-series model (Bi-LSTM + VAE) achieved 92.01% accuracy, while the MRI-based 3D CNN reached 88.73% accuracy for NHY scale classification. However, when both

modalities were integrated, the model's performance significantly improved, reaching 94.37% accuracy, demonstrating the complementary nature of time-series and MRI features. Furthermore, extending the time-series window to four time steps further boosted accuracy to 96.78%, highlighting the importance of temporal depth in multimodal learning. These results indicate that MRI data enhance structural understanding, while time-series data provide valuable temporal insights, making their combination a powerful strategy for PD classification.

In addition, we analyze the proposed models in various configurations and with various time steps. It notices that increasing the number of time steps beyond four causes the model's performance to deteriorate for single tasks and multitasking (two tasks). However, when jointly trained for multitasking with three tasks, the model's performance improves steadily as the number of time steps increases. To the best of our knowledge, the proposed multitasking with various model configurations across the time steps never investigates in the context of PD with multi-modalities as shown in Table 7. All prior studies, as can be shown, have focused their analyses on a particular specific area. Because there is insufficient data and the data has significant time steps, time-series data analysis is not common in the PD.

VI. LIMITATIONS AND FUTURE WORK

Although our suggested model performs well in predicting the stage and severity of Parkinson's disease (PD), its dependence on a particular dataset—the Parkinson's

Progression Markers Initiative (PPMI)—remains a significant restriction in terms of generalizability. Despite providing high-quality multimodal data, PPMI's application in a variety of clinical contexts is limited by its lack of external validation. By using class balancing, stratified sampling, and k-fold cross-validation (SMOTE) and creating a modular architecture that can be modified to fit different datasets, we were able to lessen this. Other issues that impair prediction accuracy when modeling NHY evolution include noisy measurements, class imbalance in later phases, and missing longitudinal data. Furthermore, the preprocessing complexity is increased by combining motor, non-motor, and MRI information. Although Bi-LSTM is an excellent model of temporal patterns, it is still computationally intensive and does not provide ground truth at later stages. We intend to use deep learning-based imputation to handle missing data, integrate wearable sensor data for continuous monitoring, validate the model on external datasets like PDBP, and investigate transformer-based architectures to improve generalizability and better capture long-range dependencies. We aim to look at dynamic time-step selection methods that are led by attention or uncertainty estimation in future work. This will allow us to represent time in a way that is particular to each patient. Adding probabilistic calibration loss functions and post-hoc refining procedures might also make the model easier to understand and ready to use in clinical settings.

VII. CONCLUSION

We present a multimodal multitask deep learning architecture based on longitudinal time-series data with extra baseline patient information to precisely forecast Parkinson's disease (PD) progression and related depression. This method presents a comprehensive picture of disease progression by using multimodal data from the Parkinson's Progression Markers Initiative (PPMI) dataset, together with neuroimaging (MRI), motor function scores, and non-motor function assessments. The suggested technique shows better performance than past models by using deep learning-based feature extraction and classification. The model uses a hybrid deep learning framework whereby extracted numerical features from motor and non-motor function data are processed using a variational autoencoder (VAE), thereby improving feature representation and lowering redundant or unnecessary information. Concurrent with this, deep MRI features are learnt using a 3D Convolutional Neural Network (3D-CNN), therefore capturing complex structural changes in the brain linked with PD. These feature sets are then merged and input into a Bidirectional Long Short-Term Memory (Bi-LSTM) network which effectively represents temporal relationships inside the time-series data, hence boosting the accuracy of illness severity and depression level forecasts. By offering better classification accuracy among the suggested approaches, the variational autoencoder (VAE) with 3D CNN MRI feature fusion surpasses other feature selection strategies, like SULOV recursive feature selection. Using two-time

steps, the suggested model obtains the maximum cross-valuation (CV) accuracy of 96.78% and testing accuracy of 94.37% in projecting the NHY scale for PD severity classification according experimental findings. Furthermore, improving generalization is the multitask learning approach, whereby the model simultaneously forecasts many clinical parameters (PD stage, depression status, and depression severity). Testing accuracy consistently increases throughout 2-time step (86.29%), 4-time step (87.12%), and 6-time step (95.81%) studies. These results stress the advantages of multimodal data fusion for complete disease modeling and the important part deep learning architectures play in clinical decision support. Adding temporal smoothing to the model made it much more calibrated and less variable in its predictions, all without changing the network's design. Our results show that while larger temporal histories (such 6-time step) might give small improvements in some measures, the 4-time step configuration is a more stable and generalizable solution across all clinical evaluation criteria.

TABLE 9. Acronyms and abbreviations.

Acronym	Definition
PD	Parkinson's Disease
NHY	Hoehn and Yahr Scale
PPMI	Parkinson's Progression Markers Initiative
NMS	Non-Motor Symptoms
MRI	Magnetic Resonance Imaging
UPDRS	Unified Parkinson's Disease Rating Scale
GDS	Geriatric Depression Scale
HAM-D (HDRS)	Hamilton Depression Rating Scale
BDI	Beck Depression Inventory
MADRS	Montgomery-Åsberg Depression Rating Scale
HADS	Hospital Anxiety and Depression Scale
Bi-LSTM	Bidirectional Long Short-Term Memory
3D-CNN	3D Convolutional Neural Network
VAE	Variational Autoencoder
SMOTE	Synthetic Minority Over-sampling Technique
KL Divergence	Kullback-Leibler Divergence
ELBO	Evidence Lower Bound
CV	Cross-Validation
OPT	Bayesian Optimization
FS	Feature Selection
SULOV	Searching for Uncorrelated List of Variables
ML	Machine Learning
AI	Artificial Intelligence

CONFLICTS OF INTEREST

The authors declare that we have no conflict of interest.

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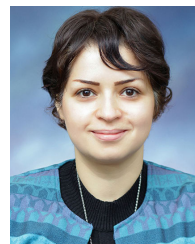
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